APPLICATION OF COLOR TREND ANALYSIS AND WCAG-COMPLIANT COLOR RECOMMENDATION STRATEGIES FOR UI/UX DESIGN.

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DECLARATION

I declare that the content of this undergraduate thesis Application of color trend analysis and WCAG-compliant color recommendation strategies for UI/UX design is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for any other degree in any university or institution of higher learning.

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ABSTRACT

Colors play an important role in the designing of user interfaces. Colors affect the perception of users as well as their emotional response and make the use of applications more efficient and enjoyable. The choice of proper color palettes has become crucial in today's interface design as it is vital for the accessibility, usability, and inclusiveness. However, there are still many designers who base their choices on personal opinion and preference that leads to low-quality designs and poor accessibility compliance. Moreover existing color selection tools frequently emphasize either visual attractiveness or accessibility, but rarely integrate both aspects into a unified and framework. This research investigates color trend analysis and WCAG-compliant color recommendation strategies for UI/UX design. This study aims to examine historical patterns of color usage in digital interfaces, evaluate the impact of color on user experience and usability, assess accessibility requirements in accordance with the Web Content Accessibility Guidelines (WCAG 2.1), and explore the application of intelligent recommendation methods for color selection. A quantitative research approach will be adopted, using secondary data collected from UI/UX design repositories, digital design platforms, and publicly available datasets. The collected data will be analyzed to identify dominant color patterns, industry-specific trends, and accessibility-related characteristics. Based on the findings a web-based Color Trend Analysis and Recommendation System will be developed using React for the frontend, Node.js for the backend, and a database for storing color trend information. The system will examine past uses of colors, check combinations based on WCAG guidelines for accessibility and make recommendations for successful color combinations based on data analysis. The system is designed to aid UI/UX designers in selecting color combinations which are both aesthetically pleasing and accessible to various users. The expected outcome of this research is a framework that combines color trend analysis, accessibility evaluation and intelligent recommendation strategies to support informed design decisions. The study contributes to the fields of Human Computer Interaction , UI/UX design, and accessibility by encouraging the development of digital products that are aesthetically effective, user-friendly, and inclusive.

Keywords: UI/UX Design, Color Trend Analysis , Accessibility Guidelines (WCAG 2.1) , Accessibility Color Recommendation , Design Strategy, Color Analytics.

Table of Contents

DECLARATION iii

ACKNOWLEDGEMENT iv

ABSTRACT v

Table of Contents vi

List of Figures xi

List of Tables xiii

List of Abbreviations xiv

1.INTRODUCTION 1

1.1 Chapter overview 1

1.2 Background of the study 1

1.3 Problem statement 2

1.3.1 General problem 3

1.3.2 Specific problem 3

1.4 Research question 3

1.5 Research motivation 3

1.6 Research aim 4

1.7 Rich picture of the proposed solution 5

1.8 Resource requirements 5

1.8.1 Hardware requirement 5

1.8.2 Software requirement 6

1.9 Project scope 6

1.10 Chapter summary 6

2.OBJECTIVES 7

2.1 Chapter overview 7

2.2 Research aim 7

2.3 General objective 7

2.4 Specific objectives 7
2.5 Research questions 7
2.6 Objective-to-research-question mapping 8
2.7 Chapter summary 8
3. LITERATURE REVIEW 9
3.1 Chapter overview 9
3.2 Conceptual map 9
3.3 Importance of color in ui/ux design 9
3.4 Color psychology and user experience 11
3.5 Evolution of color trends in digital design 11
3.6 Cultural influence on color preferences 12
3.7 Accessibility and WCAG compliance 13
3.8 Data-driven color analysis 15
3.9 Artificial intelligence and color recommendation systems 15
3.10 Existing systems/frameworks 16
3.11 Critical analysis 16
3.12 Research gap 17
3.13 Chapter summary 18
4. METHODOLOGY 19
4.1 Introduction. 19
4.2 Research design 19
4.3 Data collection 20
4.4 Data preprocessing 21
4.5 Research methodology execution workflow 22
4.6 Project management methodology 23
4.7 Tools & technologies 24
4.8 System development process 24

4.9 Project timeline 25
4.10 System requirement specification 26
4.10.1 Stakeholder analysis 26
4.10.2 Operationalization process 27
4.10.3 System/model analysis 29
4.10.4 Proposed system architecture diagram 36
4.10.5 Functional and non-functional requirements 37
4.10.6 System requirements summary 39
4.11 Implementation / designing 39
4.11.1 Pseudocode / Flow logic / Algorithms/Models 41
4.11.2 Framework workflow series 51
4.11.3 Technology selection justification 52
4.11.4 Programming language 53
4.11.5 Libraries used 53
4.11.6 Backend and front-end frameworks used 53
4.12 Testing and evaluation methodology 54
4.12.1 Test plan and test cases 55
4.12.2 Nonfunctional testing 60
4.12.3 Functional testing 61
4.12.4 Testing / Evaluation workflow 62
4.12.5 Review of testing strategies used (research suitability) 63
4.12.6 Testing methodology summary 63
4.13 Ethical considerations 64
4.14 Chapter summary 64
5. RESULTS 65
5.1 Chapter overview 65
5.2 Dynamic WCAG criteria evaluation results 65

5.3 Criteria-aware text color recommendation results 65
5.4 WCAG-synchronized analysis dashboard results 65
5.5 Trend and accessibility aggregation results 66
5.6 K-Means clustering results 66
5.7 Functional testing results 66
5.8 Non-functional testing results 67
5.9 Production build and deployment validation results 68
5.10 Results summary 68
5.11 Key implementation attempts (contribution-focused) 68
5.11.1 Attempt 1: Dynamic WCAG criteria engine (AA/AAA, normal/large) 68
5.11.2 Attempt 2: Criteria-aware text color recommendation 72
5.11.3 Attempt 3: WCAG-synchronized analysis dashboard 74
5.11.4 Attempt 4: Trend + Accessibility aggregation model 76
5.12 Implementation attempts Summary 80
6.DISCUSSION AND CONCLUSIONS 81
6.1 Chapter overview 81
6.2 Discussion of color trend and data analysis findings 81
6.3 Discussion of K-means clustering findings 81
6.4 Discussion of WCAG accessibility findings 81
6.5 Discussion of color recommendation findings 82
6.6 Discussion of dashboard and website audit findings 82
6.7 Discussion of testing and evaluation findings 82
6.8 Achievement of research objectives 83
6.9 Research contribution 84
6.10 Concluding remarks 84
6.10.1 Accomplishment of the research objectives 84
6.10.2 Problems encountered 85

6.10.3 Limitations 85
6.10.4 Self-reflection 86
6.10.5 Business insight of the idea 87
6.10.6 Future recommendations 87
6.11 Conclusion 88
6.12 Chapter summary 88
REFERENCES 89
Appendix – A : Additional System Diagrams 92
Appendix – B : System Screenshots 94
Appendix – C : Supervisory meeting minutes 102

List of Figures

Figure 1 : Conceptual map 9
Figure 2 - Use case diagram 29
Figure 3 - Class diagram 31
Figure 4 - Activity diagram 32
Figure 5 - Sequence diagram 01 33
Figure 6 - Sequence diagram 02 34
Figure 7 - Sequence diagram 03 35
Figure 8 :Proposed system architecture diagram 36
Figure 9 - Flow chart 50
Figure 10 - Block diagram A 51
Figure 11 - Block diagram B 51
Figure 12 - Block diagram C 51
Figure 13 : Testing/evaluation workflow 62
Figure 14 : Recommendation page 69
Figure 15 : Recommendation page 70
Figure 16 : Color picker 72
Figure 17 : Recommendation results 73
Figure 18 : Analysis page 75
Figure 19 :Analysis page 75
Figure 20:Accessibility page 76
Figure 21 :Color frequency chart 77
Figure 22:Trend vs Accessibility panel 78
Figure 23:Colors sorted by frequency 79
Figure 24 :Deployment Diagram 92
Figure 25 : State Chart 93
Figure 26 :system screenshot 94
Figure 27 : System screenshot 94
Figure 28 : Dataset page 95
Figure 29 : Dataset page 96
Figure 30 : Audit url 97
Figure 31 : Improvement and recommendation (audit url) 98

Figure 32 : Output of audit url 99
Figure 33 : Output of audit url 100
Figure 34 : Output of audit url 101

List of Tables

Table 1 : List of abbreviations xiv
Table 2: Hardware requirements 5
Table 3 : Software requirement 6
Table 4 : Project scope 6
Table 5 :Objective-to-research-question mapping 8
Table 6 : Existing systems 16
Table 7 : Research methodology execution workflow 22
Table 8 : Project management methodology 23
Table 9 : Project timeline 25
Table 10 : Stakeholder analysis 26
Table 11 : Operationalization process 27
Table 12 : Use case specification 30
Table 13 : Functional requirement 37
Table 14 : Non-functional requirements 38
Table 15 : Technology selection justification 52
Table 16 : Programming languages 53
Table 17: Libraries used 53
Table 18 : Backend and front-end frameworks used 53
Table 19 :Test plan and test cases 55
Table 20 :Summary of functional testing outcomes 66
Table 21 : AA/AAA Decision chart 80
Table 22 : Achievement of research objectives 83

List of Abbreviations

Table 1 : List of abbreviations

Abbreviation	**Meaning**
UI	User Interface
UX	User Experience
WCAG	Web Content Accessibility Guidelines
AI	Artificial Intelligence
ML	Machine Learning
DSRM	Design Science Research Methodology
HEX	Hexadecimal
RGB	Red, Green, Blue
AA	Level AA
AAA	Level AAA
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
HCI	Human-Computer Interaction
URL	Uniform Resource Locator

.

1.INTRODUCTION

1.1 Chapter overview

This chapter presents the research by presenting the problem background, problem statement, research questions, motivation, aim, objectives, proposed solution, project scope, required resources, and chapter summary. It provides the foundation for the study and explains why research on color trend analysis and WCAG-compliant color recommendation strategies is important for modern UI/UX design.

1.2 Background of the study

The rapid development of digital technologies has increased the significance of User Interface (UI) and User Experience (UX) design in determining the effectiveness and success of websites, mobile applications and software systems. Color is one of the most influential factors of user interaction and perception among the different visual elements of digital interfaces. Appropriate use of colors in digital interfaces helps improve readability, supports easier navigation, creates emotional connections with users, strengthens brand identity, and increases overall user satisfaction[1].

Over the last decade, UI/UX design has moved from static visual interfaces to more dynamic, adaptive, and personalized digital experiences. Thus, the use of color in digital products has also significantly changed, driven by technological advances, new design trends, accessibility needs, and changing user preferences [2] [3]. Today, designers are tasked with creating interfaces that are visually appealing, functional, accessible and user-centric. Although these requirements, many designers still rely heavily on personal experience and subjective judgement in selecting color combinations, which can cause inconsistent design results and lower usability [4] [5].

Different aspects of color use in digital systems have been studied in previous research. [6][7] have explored the relationship between color and emotional response, user behavior and perceived usability. The impact of color on website aesthetics, visual complexity, and users' initial impressions of interfaces has also been explored in research [8][5]. In addition to this, differences in color perception among cultures have been recognized as another important factor influencing preferences and reliability of users in different contexts [9]. In addition to this, guidelines like Web Content Accessibility Guidelines (WCAG) have been established to

maintain usability and accessibility for people with different abilities and visual impairments [10] [11].

The recent advancements in artificial intelligence, machine learning, and intelligent recommendation methods have further enhanced the design process using digital media. The use of data and predictive methods allows designers to understand the behavior of users and make better decisions about design [12][13][14]. There is potential in such technology to develop intelligent systems that can analyze design trends and recommend appropriate color combinations.

Despite valuable contributions from previous works, current research is mainly focused on individual aspects including aesthetics, accessibility, color psychology, or prediction techniques. There is an absence of integration of color trend analysis, accessibility assessment, and intelligent recommendation approaches into one approach. That is why the objective of this study will be the investigation of color trend analysis and WCAG compliant color recommendations for UI/UX design.

The literature review of existing works concerning the significance of color in UI/UX design, color psychology, design trends, accessibility requirements, data analysis techniques, and AI recommendations will be performed in this study. These results will serve as a basis for the current work and help identify the research gap justifying the necessity to develop a Color Trend Analysis and WCAG Compliant Color Recommendation System.

1.3 Problem statement

The use of colors in UI/UX design is significant since they affect the aspects of readability, usability, accessibility, users' perception, and users' satisfaction. However, choosing appropriate color palettes is still a difficult task for most designers, who normally use their personal experiences and perceptions during the process of color selection.

The previous researches have been conducted in the fields of color psychology, website aesthetics, accessibility, and artificial intelligence-based recommendations. Nevertheless, most previous studies concentrate either on color psychology, website aesthetics, accessibility, or artificial intelligence-based recommendations but not on the integration of all the above factors in one study. Studies on color trends are concerned only with visual preferences and design trends, while studies on accessibility mostly deal with standard requirements such as WCAG.

Consequently, there is a lack of approaches that integrate historical color trends analysis, WCAG accessibility analysis, and intelligent recommendation techniques. Hence, there is a need for a system that will be able to offer color recommendations based on trends and taking into consideration their accessibility.

1.3.1 General problem

As digital products become more dependent on visual design components to enhance the experience of the users, improper choice of color could impact their usability, readability, accessibility, and overall satisfaction with the product.

1.3.2 Specific problem

The existing practices for selecting colors among UI/UX designers involve personal preference and subjective evaluation while there have been few researches on color trends analysis, accessibility issues, or intelligent recommendations separately.

There is a lack of studies that consider all these components within one system.

Hence, the need for Color Trend Analysis and WCAG-Compliant Color Recommendation System arises.

1.4 Research question

Main Research Questions

How can the use of color trends analysis and WCAG compliant recommendation help in choosing colors for UI/UX design?

Secondary Questions

RQ1: How does color trend affect UI/UX design?

RQ2: How can data from past colors aid in design decision-making?

RQ3: How can WCAG standards be incorporated into the recommendation system?

RQ4: How effective is the recommended recommendation system?

1.5 Research motivation

Choosing colors for UI/UX design is often based on personal taste and intuition, or popular design trends. Although this can, sometimes create attractive designs, it may also result in unsuitable color combinations, inconsistent designs, and accessibility problems for users with

visual impairments. As inclusive design becomes more important and artificial intelligence continues to support design processes, there is a growing need for a more reliable and systematic way to select colors. This research was motivated by the need to develop a color analysis and recommendation system that can help designers choose visually appealing, harmonious, and accessible color combinations for UI/UX applications.

1.6 Research aim

The major objective of this study is to conduct a color trend analysis along with developing strategies of recommending color palettes that adhere to the Web Content Accessibility Guidelines for UI/UX designs and developing an intelligent color selection system.

1.7 Rich picture of the proposed solution

Workflow:

UI/UX Dataset Collection

↓

Data Processing and Preparation

↓

Color Trend Analysis

↓

WCAG Accessibility Validation

↓

Recommendation Engine

↓

Suggested Color Palette

↓

User Interface Output

1.8 Resource requirements

1.8.1 Hardware requirement

Table 2: Hardware requirements

Hardware	Specification
Processor	Intel Core i5 or above
RAM	8GB minimum
Storage	256GB SSD
Internet	Required

1.8.2 Software requirement

Table 3 : Software requirement

Software	Purpose
React.js	Frontend
Node.js	Backend
Visual Studio Code	Development
Supabase	Database
GitHub	Version Control
Draw.io	Diagrams
Chrome Browser	Testing

1.9 Project scope

Table 4 : Project scope

In Scope	Out of Scope
Analyze UI/UX color trends	Typography analysis
Evaluate WCAG accessibility	Animation analysis
Generate color recommendations	Complete UI generation
Historical color data analysis	Graphic design tools

1.10 Chapter summary

This chapter provides an introduction to the background and the problem area of color selection in UI/UX design. The research questions, motivation, objectives, scope, proposed solution, and needed resources have been defined in this chapter.

2.OBJECTIVES

2.1 Chapter overview

This chapter defines the research aim, general objective, specific objectives, research questions and the relationship between the objectives and the research questions. The objectives guide the methodological design, system development, evaluation and interpretation of the study.

2.2 Research aim

The major aim of this study is to conduct color trend analysis and develop WCAG-based color recommendation strategies for UI/UX design through a web-based, data-driven color selection system.

2.3 General objective

To develop a web-based color trend analysis and WCAG-based color recommendation system to support accessible and data-driven UI/UX color selection.

2.4 Specific objectives

1. To determine present color trends employed in the design of UI/UX interfaces.

2. To study how the choice of colors affects usability and accessibility.

3. To create a color trend analysis system that recommends colors based on WCAG criteria.

4. To assess the efficiency of the proposed system.

2.5 Research questions

Main Research Question : How can color trend analysis and WCAG-compliant recommendation help in choosing colors for UI/UX design?

RQ1: How does color trend affect UI/UX design?

RQ2: How can data from past colors aid in design decision-making?

RQ3: How can WCAG standards be incorporated into the recommendation system?

RQ4: How effective is the proposed recommendation system?

2.6 Objective-to-research-question mapping

Table 5 :Objective-to-research-question mapping

Objective	Related Research Question
O1 – Determine UI/UX color trends	RQ1, RQ2
O2 – Study usability and accessibility effects	RQ1, RQ3
O3 – Develop WCAG-based recommendation system	RQ3
O4 – Assess system efficiency	RQ4

2.7 Chapter summary

This chapter defines the research aim and objectives that guide the study. The research questions establish the information that the methodology, system development, and evaluation are intended to address.

3. LITERATURE REVIEW

3.1 Chapter overview

In this chapter, the current literature on color usage in UI/UX design and accessibility-based recommendation systems is presented. The chapter describes the relevant literature on color psychology, color trends, accessibility criteria, intelligent recommendation methods, and data-driven methodologies employed in digital products' design. In addition to that, existing methods and technologies are evaluated in order to find out their advantages, disadvantages, and research gaps. The results obtained through the analysis of the reviewed literature serve as the theoretical basis of the proposed system.

3.2 Conceptual map

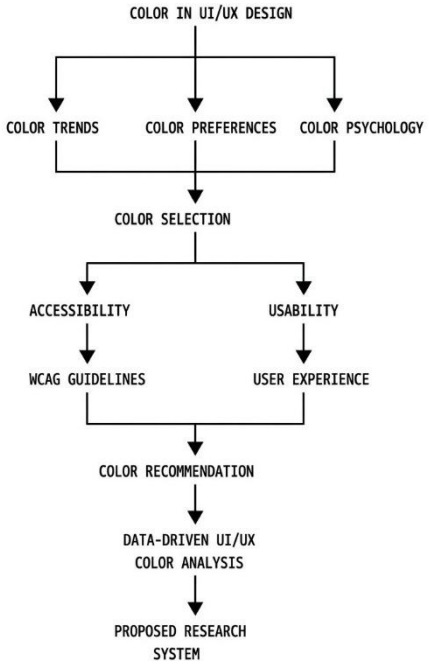


Figure 1 : Conceptual map

3.3 Importance of color in ui/ux design

Color is one of the most powerful visual elements in user interface (UI) and user experience (UX) design. It affects users' first impressions, their emotional response, readability, navigation and overall satisfaction. Good visuals and proper colors can lead to good first impressions and better user involvement [5]. According to Seifi and Moshayeri , visual appeal may increase usability through making navigation easier and decreasing the cognitive

burden of the interaction [5].Sites with good harmony, balance, and consistency in their visuals are seen as being more usable and pleasant by users [5].

Norman [1]further elaborated, design should focus on the UI/UX functionality and emotional experience, and therefore the color is an important part of human-centered design. The right choice of colors can enhance usability and create meaningful user interactions. Good design is not only about how something looks [1], says Norman, but about supporting the understanding and interaction of users.

Research has demonstrated that color has a direct influence on website appeal and user engagement Alotaibi [4] concluded that selecting effective colors improves user attention, visual attractiveness and quality of interaction.

Similarly, Peddisetty [15] stated that UI/UX design elements such as color are critical in increasing user satisfaction and improving the performance of digital products. A good UI/UX design is all about user satisfaction, usability and accessibility. Consistent color schemes and intuitive interfaces make the navigation of digital products more efficient and improve the overall user experience [15].

This increased dependency on digital products has raised the need for thoughtful color choice instead of relying on an instinctive choice. Designers have been obliged to think of usability, accessibility, and emotions along with current trends in design all at once. Color has thus become a design element because of its functional importance in digital products [2].

Color is viewed as a strategic component in digital products as it affects the attention span, branding, and user experience of users. As described by Lin [2], the role of color comes in as the first piece of visual information that users process. Color aids in developing brand identity, information hierarchy, emotion, and accessibility in digital interfaces [2]. Well - utilized color aids in information architecture by directing users' attention towards important interface elements [2]. Joo [16] said that the interacting interfaces play a big role in digital systems. He also said that interface elements are important to improve the user's interaction. Joo [16]emphasized the importance of understanding UI/UX concepts and design principles to enhance the development of interfaces and the creation of content.

According to Srimurugan [3], clarity, consistency, and ease of use are some of the major design factors influencing user satisfaction and effectiveness of task completion. An effective interface not only increases efficiency and minimizes errors but creates satisfaction and

engagement as well[3].User-centered design is important in designing computer interfaces that increase efficiency and engagement[3].

3.4 Color psychology and user experience

Color psychology is defined as the psychological study of the effect of colors on human behavior, perception, and mood. It is an important element in user interface design.

There was an extensive research by Ou et al. [6], which examined color emotions and revealed the dimensions of color perception to include warmth, activity, and emotional dominance.

Color influences emotions and perceptions of users; thus, users' interactions with digital systems are influenced by colors [1].According to Norman [1], emotional design is very significant as far as users' perceptions of products are concerned. Emotional responses are usually positive, leading to satisfaction. In this regard, color choice must be congruent with the desired emotion.

Color of products affects both user experience (UX) and usability according to the study conducted by Paschoarelli et al [7]. From the research conducted, they indicated that users often consider products whose color scheme is visually appealing as easy to use even without using the product.

Moreover, Seifi and Moshayeri [5] found that color schemes play an important role in enhancing the satisfaction of users in web design. Proper color combinations can help attract more users, while wrong color schemes can result in dissatisfaction. The choice of color is highly influential in terms of emotional response and perception [1] [2].

As a result, color psychology is an important aspect of user interface design.

3.5 Evolution of color trends in digital design

Digital color trends have undergone substantial changes during the last decade. The interfaces of digital products that were designed previously were characterized by the usage of bright and **s**aturated colors, whereas the current interfaces tend to use minimalistic and adaptive color themes [2]. Digital interfaces continuously evolve according to technological advancements and changing user requirements, resulting in changes in design styles and visual elements such as color usage [16].

Lin [2]defined several periods of color trends development from 2015 until 2025. At the beginning of the period, digital products tended to use flat design along with vivid color combinations. The improvement of mobile technology encouraged the introduction of more natural and soft color combinations for better readability and reduction of visual strain. Lin [2]outlined four main color trends in digital products design: Digital Native Aesthetics, Natural Healing Tones, High-End Neutral Sensibility, and Bold Contrast Collisions. Trends in colors of today's modern era are affected by technology, social media networks, and behavioral changes in users [2].

The adoption of the smartphone further affected color design trends. As noted by Ebner et al., [17], the Human Interface Design Guidelines from Apple promoted the implementation of good layout design along with the use of well-chosen color schemes. This had an effect on web and mobile applications design in different industries.

The technique of color forecasting has been in use in industries like fashion and textiles for quite some time now. Cassidy [18] pointed out that consumer preferences and cultural trends have a large impact on future color choices. The same kind of forecast has now come into use in digital design.

As reported in recent studies conducted by Srimurugan [3], it becomes evident that contemporary UX/UI design practices are gradually incorporating trend analysis to study user expectations. Thus, analysis of previous color trends offers great insight into color selection for future products. "The world of digital product interfaces is ever-changing. According to Srimurugan (2025), interface architectures have 'evolved considerably since the last few decades as a response to the changing technological scenario and user needs.' With the shift of interactions from command line-based interfaces to virtual environments, it is essential for the interface requirements to evolve accordingly in order to be responsive to technological changes. The evolving nature of interface systems supports the case for mapping a decade's worth of color information (2015-2025)"[3].

3.6 Cultural influence on color preferences

Color preferences differ from one culture to another and even between generations and societies. It is essential to know about such differences in order to create a digital product for international use.

In their research, Cyr et al. [9] studied how people in various cultures perceive colors and discovered major differences in color preferences.

As stated by Cassidy [18], consumer color preferences develop depending on the cultural and social evolution. The same color can have completely different associations in two cultures. For instance, a color that represents happiness in one culture can have nothing to do with it in another culture.

Hong and Kim's research into personal color analysis [19] also showed that people react to colors in different ways based on their own individuality. This implies that the development of future color recommendation systems will need to include personalization algorithms to improve their performance. According to Lin [2] ,color trends are influenced by: AI technology, Social media, Post-pandemic behavior, Generational preferences.

3.7 Accessibility and WCAG compliance

Accessibility is an essential attribute for today's interface design. Digital products have to be accessible to all users, regardless of whether they suffer from visual impairment or color blindness. Web Content Accessibility Guidelines (WCAG 2.1) [10]set internationally acknowledged norms of accessibility. It is necessary to achieve a certain ratio of contrast to make sure that the content will be visible and readable.

According to WCAG 2.1 [10], color should not be relied on as the sole method of communication through digital interfaces. WCAG 2.1 suggests adequate contrast between text and background to enhance readability [10].WCAG 2.1 suggests improved contrast requirements to increase accessibility for visually impaired and reading-challenged users, thus making the digital content visible and easy to read [10].WCAG 2.1 states that the digital content must stay readable under various conditions and on all kinds of devices [10].Usability is enhanced not only for those with disabilities but for all users through accessibility principles [10].

Shah [11] stressed the significance of incorporating WCAG principles into the design systems of today. Increasingly, many firms have started employing an accessibility-first mindset to ensure compliance with regulatory requirements. Standards for accessibility must be included in design systems to help create an inclusive and user-oriented experience [11].With the help of WCAG 2.1, some new accessibility requirements were created to cater to the modern

digital environment [11].Compliance with accessibility involves more than just guideline adherence; it requires an assessment, implementation, and testing process as well [11].

Research conducted by Shiraiwa et al. [20] found out that text visibility depends greatly on the combination of colors used for text and background. Color combinations of texts and backgrounds have a significant role to play in enhancing the readability and information communication within digital interfaces [20]. Luminance difference in addition to the choice of colors determines visibility within digital interfaces [20].Sufficient luminance differences enhance readability [20].In addition, Vanderschantz et al. [21]concluded that good color contrast leads to good reading performance. Color choices for text and backgrounds have a significant effect on readability in the digital environment [21].

Color Guard is an automated system developed by Balasooriya and Wickramarathne [22] that checks if the system conforms to the Color Design guidelines. Such research demonstrates the growing need for tools that evaluate accessibility automatically. Even though trend colors can make an interface more visually appealing, some combinations might be difficult to read and fail accessibility criteria[2].Trending color combinations have to conform to the requirements of accessibility and usability to create an inclusive experience for users [2].Accessibility has become one of the key criteria in modern interface design, especially with regard to people with disabilities [3]."In order to move from an intuitive approach to color optimization in interfaces to an engineering process, the interface evaluations must account for internationally accepted compliance standards. An example of this framework is demonstrated by Srimurugan (2025) through his implementation of 'Accessibility & Usability Score' calculated according to heuristics defined by Nielsen and WCAG in the context of system testing. This sets an academic standard for measuring the effectiveness of the interface and proves that modern color recommendation systems must not only follow the cycle of aesthetic preferences, but also check and calculate their output against WCAG requirements"[3]. Accessibility-oriented design processes, such as the use of high contrast colors, can contribute toward building accessible interfaces which accommodate people with various capabilities[15].

However, despite the advancements made, many designers continue to have difficulties in evaluating accessibility in their designs. Consequently, WCAG validation within color suggestion systems presents a promising area for research.

3.8 Data-driven color analysis

Due to the rising availability of design data in the digital environment, data-driven approaches can be used for color analysis.

According to Han et al. [23], data mining techniques have the ability to uncover trends in big data sets. Data mining offers unique possibilities for analysis of color utilization on digital products.

Lin [2] highlighted that trend analysis of the past may give insights into major color patterns that emerged in particular industries during particular time periods. This could help make evidence-based design decisions, not subjective ones.

Yeboah et al. [12]introduced an intelligent approach to color processing that integrates analytical tools with design optimization techniques. It shows how analysis of data can enhance color selection process.

The suggested research takes a similar view on the issue and is going to analyze color data obtained from UI/UX designs.

3.9 Artificial intelligence and color recommendation systems

In recent times, AI is becoming popular in UI/UX design. AI provides various technologies through which automated process of analysis, prediction and recommendation can be performed, providing help to the designer in decision making.

Dou et al. [13] proposed an AI framework called Webthetics which is capable of assessing web page aesthetics. The results revealed that machine learning models are capable of predicting aesthetic quality using design attributes.

Nurfadillah and Andarsyah [14] discussed the machine learning techniques used in color prediction and recommendation. The study revealed that algorithms like K-Means clustering and Random Forest can effectively create appropriate color recommendation.

Durgam et al. [24] explained the increasing use of AI-created design components and intelligent design systems. AI can help designers to create layouts, images, and color combinations using user requirements.

These examples suggest that there are various opportunities with respect to color recommendation system using AI.

3.10 Existing systems/frameworks

Table 6 : Existing systems

Study	Method	Strength	Limitation
Webthetics	Deep Learning	Aesthetic analysis	No accessibility
Color Guard	WCAG validation	Accessibility	No trend analysis
Personal Color Analysis App	Recommendation	Personalization	Limited scope

3.11 Critical analysis

Earlier works have extensively studied the effect of colors in UI/UX design from various perspectives such as user experience, psychological responses of users, accessibility, and technology. Norman [1] and Peddisetty [15] highlighted that color plays an important role in usability and user satisfaction whereas Ou et al. [6] and Paschoarelli et al. [7] have discussed the psychological effects of colors on users' perception and feelings. This shows that colors play an important role in enhancing user interaction and satisfaction.

Various authors have studied the evolution of color trends as well as the application of these trends in digital designs. Lin [2] emphasized the impact of technology and change in user preferences in color trends, whereas Cassidy [18] focused on various ways of predicting future color trends. Although the above literature gives us insights into color trend formation, none of these studies take into account accessibility issues or automatic generation techniques.

The study of accessibility has also been extensively considered in prior works. WCAG 2.1 [10], Shah [11], Shiraiwa et al. [20], and Vanderschantz et al. [21] stressed the significance of readable content, contrast ratio, and inclusive design. Nevertheless, current research on accessibility is mostly concentrated on evaluation and checking rather than direct inclusion of accessibility criteria in recommendation systems.

Due to recent progress in machine learning and data-oriented techniques, intelligent design systems can be now developed using novel methods. In particular, Dou et al. [13] used machine learning techniques for aesthetic assessment, whereas Nurfadillah and Andarsyah [14] used prediction algorithms for color suggestion. Similarly, Yeboah et al. [12] presented intelligent color processing systems. Despite the fact that all the discussed systems increase automation in designing processes, there is no system capable of evaluating both historical trends and accessibility at once.

In general, prior studies have focused on various aspects of color design, including aesthetic, psychological, accessibility, and artificial intelligence. Nonetheless, few studies have tried to integrate these aspects into an integrated system to examine color trends and provide accessibility-related suggestions for UI/UX design.

3.12 Research gap

Despite previous studies conducted on the topics of color psychology, color trends, accessibility standards, and AI recommendation systems, there are certain limitations that should be highlighted. First of all, all studies are conducted separately from each other, that is why the integrated approach is not presented yet. The majority of existing works about color trends only analyze visual preference and design evolution but not accessibility standards.

1.Previous research lacks a combination of historical analysis of colors and WCAG compliance evaluation.

2.Color recommendation tools are mostly geared towards aesthetics and not WCAG compliance**.**

3.Traits such as color trend analysis, accessibility evaluation, and intelligent recommendation are barely found together in existing literature.

4.Adequate systems that help UI/UX designers by providing trend-based and WCAG compliant color recommendations are scarce.

Hence, this paper aims at developing a Color Trend Analysis and WCAG Compliant Color Recommendation System that integrates the three concepts discussed above.

3.13 Chapter summary

This chapter provides an overview of the existing literature on color usage in UI/UX design, color psychology, color trends, accessibility criteria, data analytics, and AI-based recommendation systems. Previous literature proves the role of color in enhancing usability, users' emotional response, accessibility, and satisfaction. Nevertheless, previous research mostly focuses on separate aspects. The analysis and research gap discussion reveal the necessity to create an integrated approach combining color trend analysis and recommendation approaches meeting accessibility criteria. This knowledge serves as a theoretical basis for the proposed research and will be elaborated in the methodology discussed in the next chapter.

4. METHODOLOGY

4.1 Introduction.

The following is the methodology used in this research that is entitled “Application of Color Trend Analysis and WCAG Compliant Color Recommendation Strategies for UI/UX Design.” The methodology describes how the research was conducted and gives information on how data was collected, what technologies were used, how the system was developed and what ethical issues were considered during the research. The objective of this methodology is to develop a systematic way of analyzing color trends and developing a color recommendation system.

4.2 Research design

This study uses a quantitative research design supported by the Design Science Research Methodology (DSRM).

The quantitative approach is used to collect and analyze historical color data from digital interfaces, such as websites and mobile applications. The analysis focuses on color frequency, historical color trends, color values, contrast ratios, and WCAG compliance results. In addition, K-Means clustering is applied to group similar color combinations within the dataset.

DSRM provides the overall structure for designing, developing, and evaluating the proposed Color Trend Analysis and WCAG-Compliant Color Recommendation System. The study follows key DSRM stages: problem identification, review of existing solutions and research gaps, objective definition, system design and development, data management and analysis, and system evaluation through WCAG-based contrast assessment and software testing.The literature review serves as the theoretical foundation of the study. It is used to identify relevant concepts, existing systems, and research gaps that justify the development of the proposed system.

The research process consists of the following stages:

- Literature review and problem identification
- Data collection
- Data preprocessing

- Color trend analysis
- Accessibility evaluation
- Recommendation generation
- System implementation
- Testing and evaluation

This research design combines theoretical understanding with practical implementation.

4.3 Data collection

The data collection process is vital since the quality of the data directly impacts system performance and accuracy of recommendations.

The research relies on both primary and secondary sources of data.

Primary data sources include:

- UI/UX datasets
- Color palette datasets
- Interface screenshots
- Publicly available design resources

Secondary data sources include:

- Research papers
- Journal articles
- Books
- Online repositories
- Accessibility guidelines

The collected dataset may contain the following attributes:

- Application name
- Primary color
- Secondary color
- Color code (HEX/RGB)

- Design category
- Accessibility score
- Trend year

The data collected will be subsequently organized and prepared for analysis.

4.4 Data preprocessing

Before analysis, collected datasets will undergo preprocessing to improve quality and consistency.

Preprocessing activities include:

- Removing duplicate records
- Handling missing values
- Standardizing color formats
- Converting RGB values into HEX values
- Organizing and categorizing color groups

This process helps improve analysis accuracy and reliability.

4.5 Research methodology execution workflow

Table 7 : Research methodology execution workflow

Research Stage	Description
Problem Identification	This project starts by looking at a real problem in UI/UX design: choosing colors that look good, follow current trends, and still remain accessible to users.
Relevance Justification	The study is relevant because good interface design needs both visual appeal and accessibility. By combining color trend analysis with WCAG checks, the system helps designers make better decisions.
Comparative Analysis and Gap Justification	The project compares color use across different years and platforms, then checks which combinations pass or fail accessibility rules. The main gap it addresses is that trend analysis and accessibility checking are often handled separately, while this system brings them together in one place.
Define and Finalize Objectives	The project objectives are to collect color data, clean and prepare it, study trends, apply K-Means clustering, test contrast ratios using WCAG 2.1, and present the results in a web-based system.
Design / Development / Data Management and Handling	The system is built using React for the frontend and Node.js with Express for the backend. Data is managed through Supabase and local JSON files, with support for uploading, updating, deleting, analyzing, and recommending color data.

Evaluation and Communication	The results are shown through charts, yearly trend summaries, platform-based analysis, WCAG pass/fail results, and color recommendations. This makes the findings easy to understand and useful for decision-making.

4.6 Project management methodology

Table 8 : Project management methodology

Project Management Methodology	Justification
Agile Scrum	Scrum fits this project well because the work is divided into several connected parts, such as color data management, trend analysis, K-Means clustering, WCAG accessibility checking, and web development. Since these parts need to be developed gradually and improved along the way, Scrum makes the process easier to manage. It also allows feedback and changes during development, which is helpful for a project like this where both the technical work and research results need to stay aligned.

4.7 Tools & technologies

Various tools and technologies will be employed in system development process. React JS will be applied in designing responsive and interactive UI. Node.js and Express JS will facilitate backend operations as well as API interactions. JavaScript will be the programming language for both frontend and backend developments. Supabase will be employed for managing and storing color data. Visual Studio Code (VS Code) will help with coding and debugging processes, while Chart.js will be utilized in visualization of analysis results. GitHub will be employed in version controlling and project management. Web Content Accessibility Guidelines (WCAG 2.1) will be utilized in evaluating accessibility and validating color recommendations.

4.8 System development process

The proposed Color Trend Analysis and WCAG-Compliant Color Recommendation System will be developed using the Agile development approach.

The development stages include:

- Requirement Analysis:
  Identify system requirements.
- System Design:
  Design system architecture and user interfaces.
- Implementation:
  Develop frontend, backend, and database components.
- Testing:
  Perform functionality and usability testing.
- Deployment:
  Deploy the completed system.

4.9 Project timeline

Table 9 : Project timeline

Activities	Feb	March	April	May	June	July	Aug	Sep	Oct	Nov
Proposal & Literature Review										
Data Collection & Preparation										
Data Analysis (Trend + Accessibility)										
System Design & Development										
Testing & Report Writing										
Final Review & Submission										

4.10 System requirement specification

This part presents the requirements of the proposed system for the study, Application of Color Trend Analysis and WCAG-Compliant Color Recommendation Strategies for UI/UX Design. It explains how the research objectives are turned into practical system features and shows why each requirement is important based on the needs of the users, the collected data, and the overall system design.

Proposed platform is a web-based tool for analyzing color trends and supporting accessible UI/UX decisions. It allows users to manage the dataset, identify color trends, evaluate WCAG accessibility, group similar colors using K-means clustering, and receive readable text-color recommendations for better contrast. It also includes an optional website auditing feature for checking real UI accessibility in practice.

4.10.1 Stakeholder analysis

Table 10 : Stakeholder analysis

Stakeholder	Role in Study/System	Key Needs	Expected Benefit
Researcher	Defines objectives and evaluates outcomes	Accurate trend and accessibility evidence	Strong empirical findings and reproducible analysis
UI/UX Designer	Uses insights for design decisions	Fast palette guidance with accessibility status	Better design quality and reduced contrast errors
Accessibility Reviewer	Verifies compliance level	Transparent WCAG pass/fail logic	Efficient compliance review
Data Curator	Maintains color records	Easy create/update/delete workflows	Higher data consistency

Technical Maintainer	Operates and supports platform	Stable APIs and fault-tolerant data handling	Lower maintenance overhead
Website Owner (optional audit)	Receives accessibility diagnostics	URL-level pass/fail summaries	Actionable remediation priorities

4.10.2 Operationalization process

Table 11 : Operationalization process

Research Objective	**Data Gathering Technique**	**Dataset Variables / Inputs**	**System Operationalization**	**Validity Justification**
Identify UI color trends across platforms and time	Structured dataset collection and curation	platform, year, primary color	Frequency and year-wise trend analysis	Directly measures trend prevalence (construct validity)
Evaluate accessibility of practical color pairs	Observational color-pair data from dataset	background, text color	WCAG contrast computation and pass/fail classification	Standardized formula ensures internal consistency
Discover dominant color families	Computational analysis of color corpus	normalized primary colors	K-means clustering with configurable k	Objective grouping improves analytical rigor
Recommend readable text color for chosen background	Interactive input testing	user-provided background HEX	Contrast comparison with black/white, best candidate output	Decision rule is explicit and explainable

Extend findings to live UI contexts (optional)	URL-based website scan	website URLs	Sampled pair extraction and WCAG summary scoring	Increases external applicability of model

Requirement validity summary:

1. Construct validity is supported because each feature directly operationalizes one research objective.
2. Internal validity is supported by consistent color normalization and fixed WCAG thresholds.
3. External validity is supported through both static dataset analysis and optional real-website auditing.
4. Content validity is supported by complete coverage from data input to recommendation output.

4.10.3 System/model analysis

4.10.3.1 Use case diagram with specifications

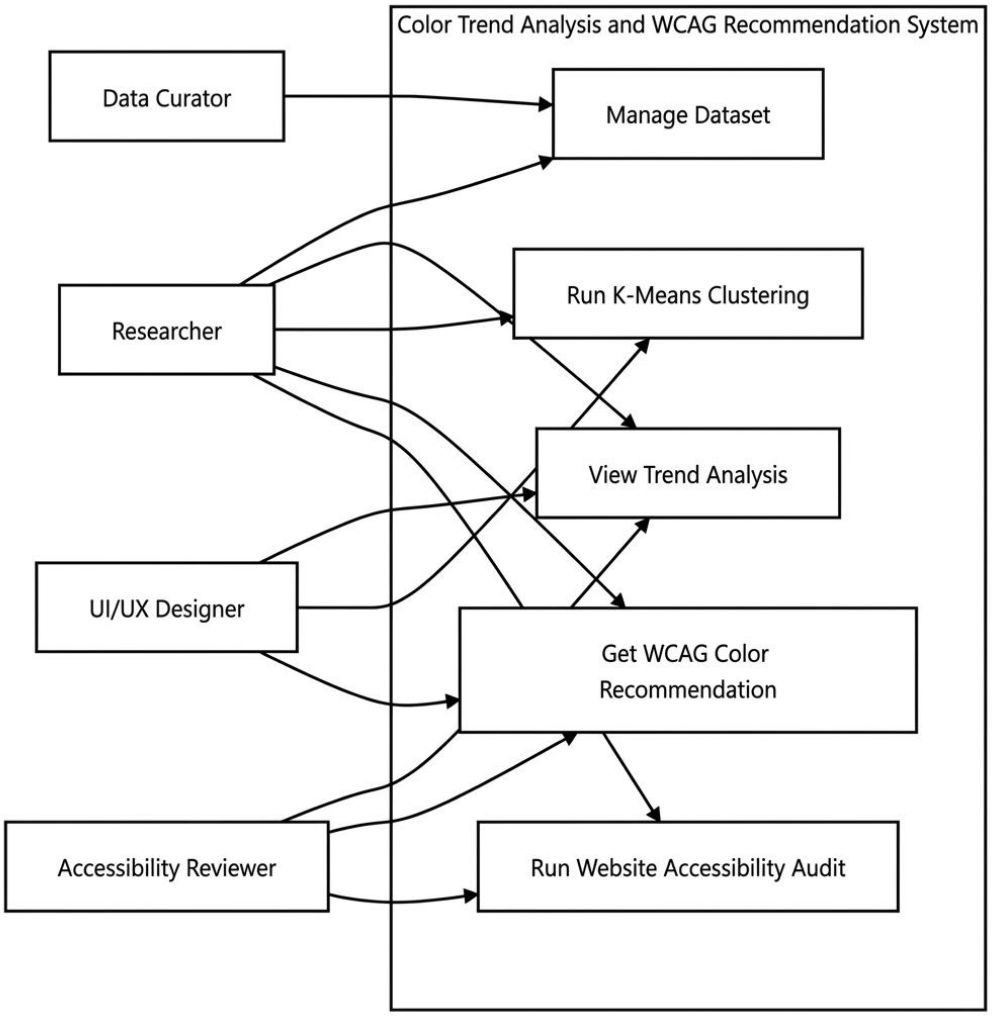


Figure 2 - Use case diagram

Use case specifications:

Table 12 : Use case specification

Use Case	**Actor**	**Brief Description**	**Main Output**
Manage Dataset	Researcher, Data Curator	Add, update, delete, and retrieve color records	Clean dataset for analysis
View Trend Analysis	Researcher, UI/UX Designer, Reviewer	Display frequency, yearly, and platform trends	Visual trend insights
Run K-Means Clustering	Researcher, UI/UX Designer	Group similar colors into clusters	Color group centroids and members
Get WCAG Color Recommendation	Researcher, UI/UX Designer, Reviewer	Recommend readable black or white text for a background color	Contrast ratio and Pass/Fail status
Run Website Accessibility Audit	Researcher, Reviewer	Evaluate website color pairs for WCAG compliance	Accessibility summary per site

4.10.3.2 Class diagram

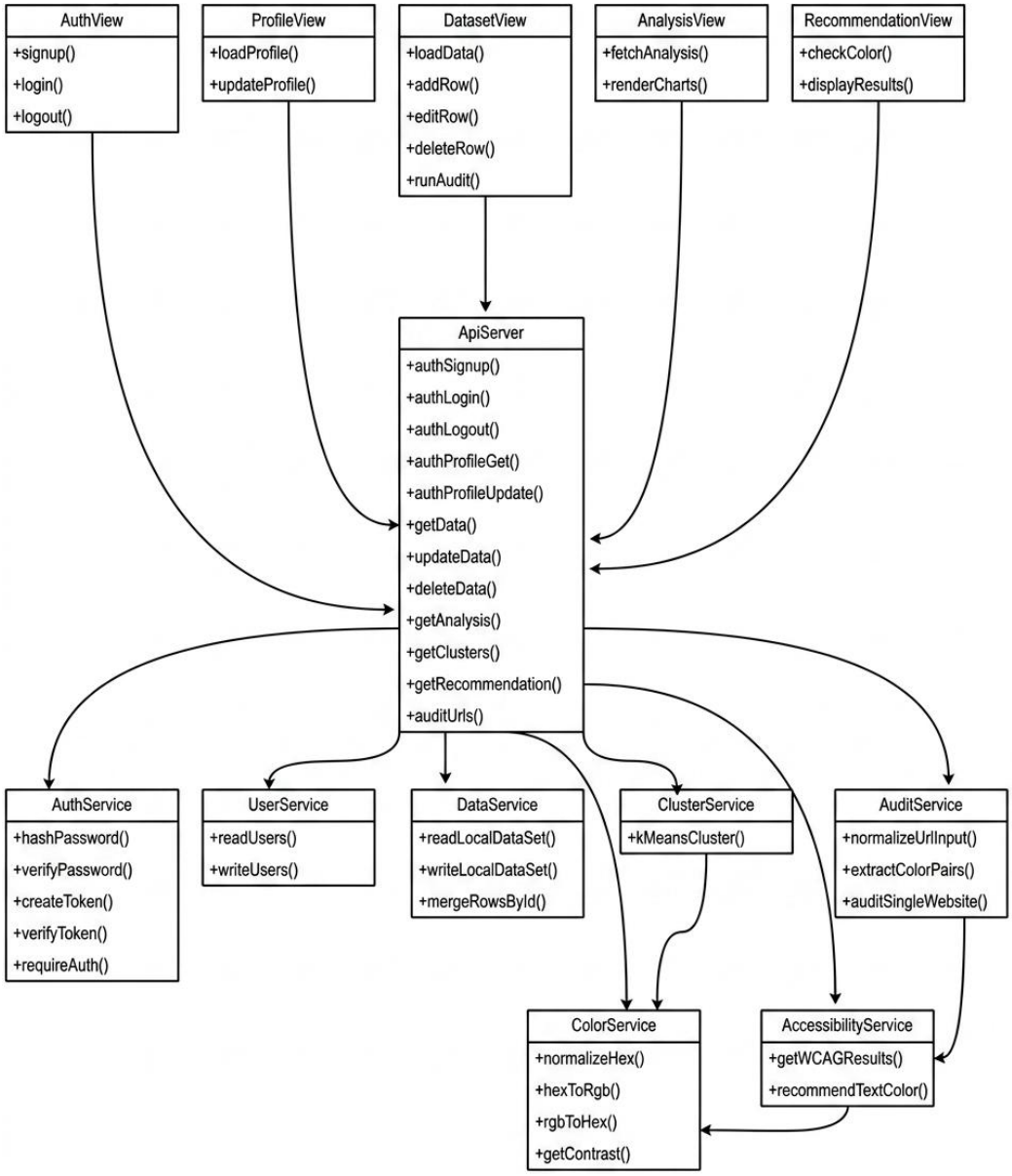


Figure 3 - Class diagram

4.10.3.3 Activity diagram

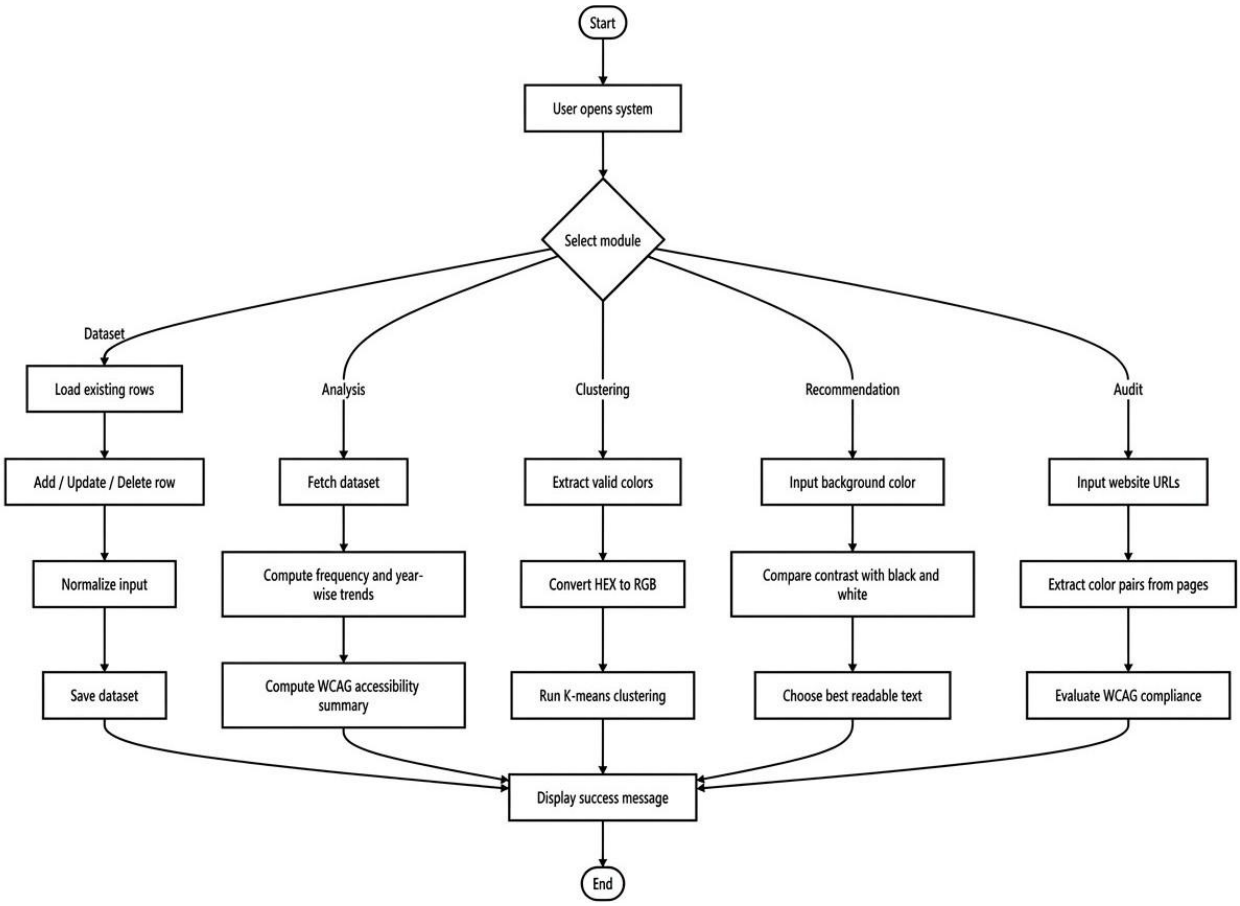


Figure 4 - Activity diagram

4.10.3.4 Sequence diagrams mapped with use cases

4.10.3.4.1 Sequence 1: Dataset management

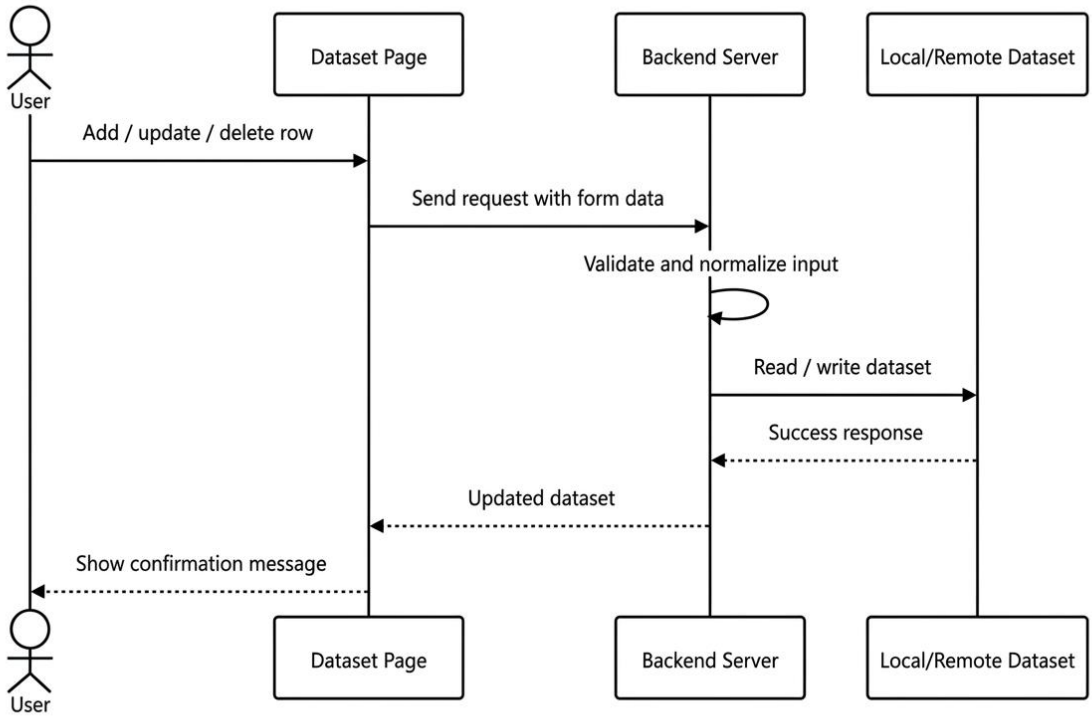


Figure 5 - Sequence diagram 01

4.10.3.4.2 Sequence 2: Trend analysis and clustering

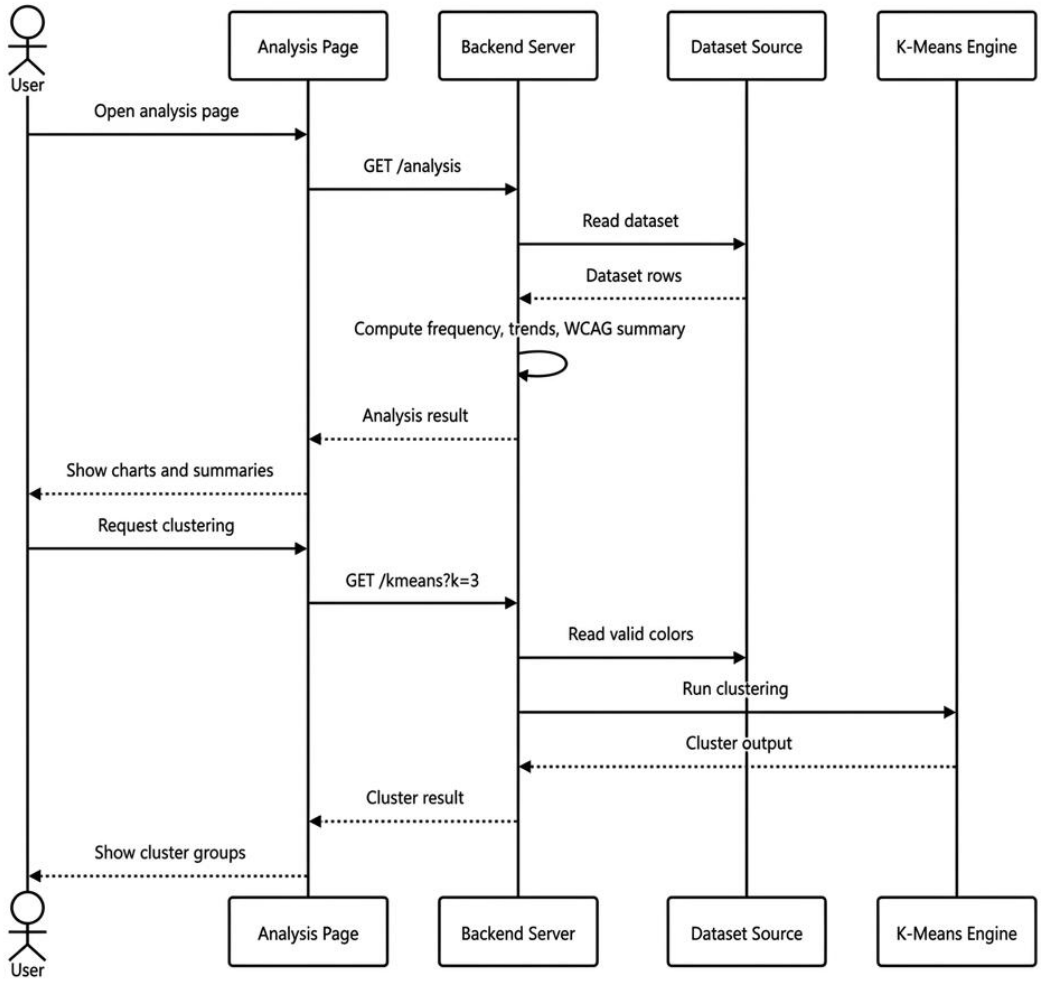


Figure 6 - Sequence diagram 02

4.10.3.4.3 Sequence 3: WCAG color recommendation

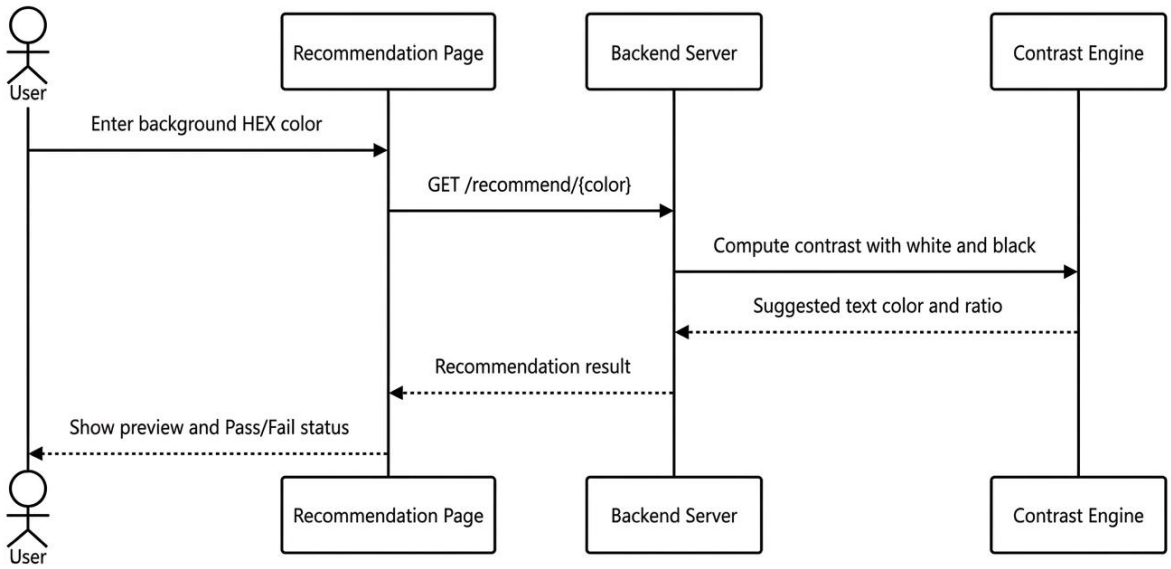


Figure 7 - Sequence diagram 03

4.10.4 Proposed system architecture diagram

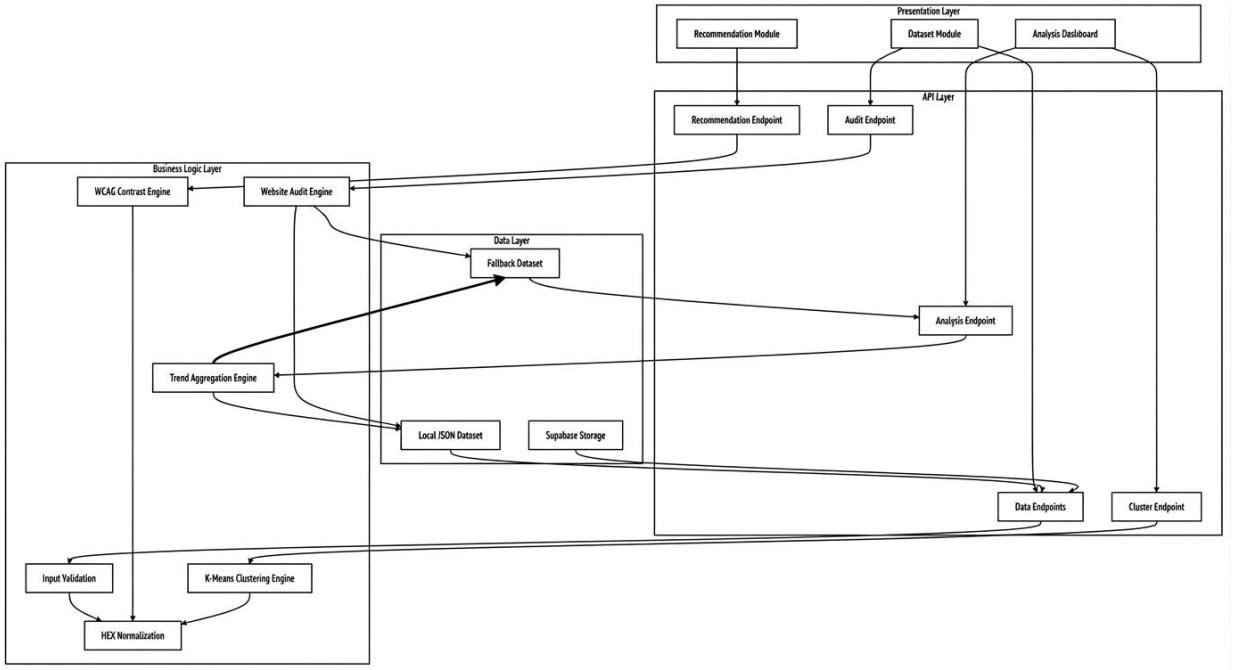


Figure 8 :Proposed system architecture diagram

4.10.5 Functional and non-functional requirements

4.10.5.1 Functional requirements:

Table 13 : Functional requirement

ID	Requirement Statement
FR-01	The system shall support dataset retrieval for analysis workflows.
FR-02	The system shall support adding new dataset rows with mandatory identifier.
FR-03	The system shall support updating existing dataset rows by identifier.
FR-04	The system shall support deleting a single row and bulk delete operations.
FR-05	The system shall normalize color values before analytical processing.
FR-06	The system shall compute color frequency distribution.
FR-07	The system shall compute year-wise and platform-wise trend summaries.
FR-08	The system shall compute WCAG(AA /AAA) pass/fail/invalid status for color pairs.
FR-09	The system shall generate K-means clustering output with configurable k.
FR-10	The system shall recommend a readable text color for a valid background HEX.
FR-11	The system shall return clear validation errors for invalid inputs.
FR-12	The system shall optionally audit submitted website URLs and produce summary metrics.

4.10.5.2 Non-functional requirements:

Table 14 : Non-functional requirements

ID	Requirement Statement
NFR-01 Performance	Typical recommendation and analysis requests should return in near real-time for normal dataset size.
NFR-02 Reliability	Service should continue with local/fallback data if remote source is unavailable.
NFR-03 Usability	The interface should present clear status feedback and readable visual summaries.
NFR-04 Maintainability	Core logic should remain modular for color, WCAG, clustering, and audit functions.
NFR-05 Scalability	URL audit should apply bounded URL and sample limits to control resource use.
NFR-06 Data Integrity	Input normalization and validation should prevent malformed analytical inputs.
NFR-07 Portability	The system should run on standard Node.js and modern browser environments.
NFR-08 Testability	Core routes and analytical functions should be verifiable through automated tests.
NFR-09 Explainability	Recommendation outcomes should be directly traceable to WCAG contrast calculations.
NFR-10 Robustness	Error handling should return meaningful failure messages without system crash.

4.10.6 System requirements summary

In this system requirement part , the full requirement specification for the proposed color trend analysis and WCAG recommendation platform was clearly defined. The needs of key stakeholders were converted into practical functional and non-functional requirements that can be measured and implemented. This chapter also showed a observe link between the research objectives, data used, and the computational methods applied.

Finally, the system models including the use case, class, activity, sequence, deployment, and architecture diagrams demonstrate that the proposed design is appropriate for building and evaluating a UI/UX tool focused on color trend insights and accessibility-aware recommendations.

4.11 Implementation / designing

Steps followed in creating the framework and designing the algorithms

The framework was developed step by step so that each research objective could be converted into a working system feature. After each stage, the implementation was checked through UI behavior and API responses to make sure the module worked as expected.

1. Problem breakdown and module definition

The overall problem was first split into clear functional modules. This helped keep the development process organized and made it easier to test each part independently. The system was structured into five major modules:

- Data management module for data upload, update, deletion, and retrieval
- Trend analytics module for frequency-based, year-wise, and platform-wise analysis
- WCAG compliance engine for contrast ratio calculation and pass/fail evaluation
- K-means clustering module for grouping colors using RGB distance
- Recommendation and audit module for text color suggestion and optional URL-level accessibility audit

2. Data modeling and normalization strategy

Since real-world data can contain inconsistent field names and incomplete values, the data model was designed to be flexible. Before any analysis, inputs were normalized to improve consistency and reliability. This included:

- HEX cleaning and standard formatting
- Handling of null, missing, or invalid values
- Use of local fallback data when remote data is unavailable

Trend analysis model : Once the dataset was clean, trend analysis was implemented. Frequency counting helped identify commonly used colors, and year-wise grouping showed how those preferences changed over time. K-means clustering was then used to group similar colors in RGB space for clearer pattern discovery.

Accessibility computation model : The accessibility module was developed using WCAG formulas for relative luminance and contrast ratio. Initially, one fixed threshold was used. Later, this was improved to support all key scenarios: - AA (normal text) - AA (large text) - AAA (normal text) - AAA (large text).This extension made the model more practical and more aligned with real accessibility evaluation.

Recommendation engine : The recommendation logic was designed to be straightforward. For a given background color, the system compares contrast with white and black text, chooses the better option, and returns both the suggestion and compliance details. This includes ratio, threshold, selected WCAG mode, and pass/fail status.

3. Service and API design

Each core feature was implemented as a separate API endpoint. This made the architecture modular and easier to maintain. It also allowed frontend components to consume services independently, which improves testing and reuse.

API and interface integration : Backend endpoints were connected to frontend pages so users could run analysis and receive recommendations through the interface. WCAG level and text-size selections in the UI were linked directly to backend calculations, ensuring the displayed results match the selected criteria.

4. Frontend interaction design

The interface was designed around three high-impact views to support research tasks and decision-making:

- Dataset view for managing curated records
- Analysis dashboard for interpreting trend and accessibility results
- Recommendation view for quick color accessibility decisions

5. Validation-focused implementation

To ensure reliable and reproducible results, logic-heavy features were implemented with deterministic rules. For example, WCAG thresholds were fixed, K-means iterations were capped, and audit sampling limits were controlled. This made the system more suitable for research evaluation and repeatable experiments.

The final stage focused on validation. Build checks, endpoint tests, and smoke tests were performed to confirm that the system works correctly and produces reliable outputs. These results were used as implementation evidence for the study.

This phased workflow improved both technical quality and research credibility, because each feature was implemented with traceable logic and verified behavior.

4.11.1 Pseudocode / Flow logic / Algorithms/Models

4.11.1.1Algorithm A: HEX Normalization

```
FUNCTION NormalizeHex(value):
  IF value is null OR value is not a string:
    RETURN null
  hex = trim(value)
  IF hex starts with "#":
    hex = hex without first character
  IF hex matches 3 hex chars:
```

```
    hex = expand each char
    # e.g., "F0A" -> "FF00AA"
  IF hex matches 6 hex chars:
    RETURN "#" + uppercase(hex)
  RETURN null
```

4.11.1.2 Algorithm B: WCAG contrast evaluation (dynamic AA/AAA, normal/large)

```
FUNCTION RelativeLuminance(hex):
  rgb = HexToRGB(hex)   # r,g,b in [0..255]
  FOR each channel c in [r,g,b]:
    c = c / 255
    IF c <= 0.03928:
      c = c / 12.92
    ELSE:
      c = ((c + 0.055) / 1.055)^2.4
  RETURN 0.2126*r + 0.7152*g + 0.0722*b



FUNCTION ContrastRatio(hex1, hex2):
  L1 = RelativeLuminance(hex1)
  L2 = RelativeLuminance(hex2)
  bright = max(L1, L2)
  dark   = min(L1, L2)
  RETURN (bright + 0.05) / (dark + 0.05)
```

```
FUNCTION GetWCAGThreshold(level, textSize):
    # Normalize inputs
    L = uppercase(trim(level))      # default "AA"
    S = lowercase(trim(textSize))   # default "normal"

    IF L not in {"AA","AAA"}: L = "AA"
    IF S in {"large","lg"}:    S = "large"
    ELSE:                      S = "normal"

    IF L == "AAA" AND S == "normal": RETURN 7.0
    IF L == "AAA" AND S == "large":  RETURN 4.5
    IF L == "AA"  AND S == "normal": RETURN 4.5
    IF L == "AA"  AND S == "large":  RETURN 3.0


FUNCTION WCAGResult(background, text, level, textSize):
    bg = NormalizeHex(background)
    fg = NormalizeHex(text)

    IF bg is null OR fg is null:
       RETURN {contrast: null, status: "invalid"}

    ratio = ContrastRatio(bg, fg)
    threshold = GetWCAGThreshold(level, textSize)
```

```
  status = "pass" IF ratio >= threshold ELSE "fail"

  RETURN {
    contrast: round(ratio, 2),
    status: status,
    threshold: threshold,
    level: normalized level,
    textSize: normalized textSize
  }
```

4.11.1.3 Algorithm C: Color recommendation strategy (criteria-aware)

```
FUNCTION RecommendTextColor(background, level, textSize):
  bg = NormalizeHex(background)
  IF bg is null:
    RETURN error "Invalid HEX"

  cWhite = ContrastRatio(bg, "#FFFFFF")
  cBlack = ContrastRatio(bg, "#000000")

  IF cWhite > cBlack:
    suggested = "#FFFFFF"
    best = cWhite
  ELSE:
    suggested = "#000000"
```

```
    best = cBlack

threshold = GetWCAGThreshold(level, textSize)
pass = (best >= threshold)

RETURN {
    input: bg,
    suggestedText: suggested,
    contrast: round(best, 2),
    status: ("Pass" if pass else "Fail"),
    level: normalized level,
    textSize: normalized textSize,
    threshold: threshold
}
```

4.11.1.4 Algorithm D: K-means color clustering

```
FUNCTION KMeansCluster(colors, requestedK):
    validHex = unique map NormalizeHex(colors), removing nulls
    points = map each validHex -> RGB point (r,g,b)

    IF points is empty:
        RETURN empty cluster result

    k = clamp(requestedK, 1, length(points))
    centroids = InitializeCentroids(points, k)
    assignments = array(length(points), -1)
    maxIter = 10

    FOR iter from 1 to maxIter:
        changed = false

        # Assignment step
        FOR i from 0 to length(points)-1:
            nearest = argmin_j Distance(points[i], centroids[j])
            IF assignments[i] != nearest:
                assignments[i] = nearest
                changed = true

        # Update step
        FOR j from 0 to k-1:
```

```
        members = all points assigned to j
        IF members not empty:
            centroids[j] = MeanRGB(members)

    IF changed == false:
        BREAK

RETURN clusters with:
    centroidRGB,
    centroidHEX,
    memberColorsHEX,
    size,
    iterationsUsed
```

4.11.1.5 Algorithm E: Trend + Accessibility aggregation (criteria-aware)

```
FUNCTION BuildAnalysis(dataset, level, textSize):
    frequency = {}
    yearWise = {}
    accessibility = {total:0, passAA:0, failAA:0, invalid:0}
    accessibilityByYear = {}
    platformAnalysis = {}

    FOR each row in dataset:
        color = NormalizeHex(row.primary)
```

```
year = ResolveYear(row) OR "Unknown"
platform = NormalizePlatform(row.platform) OR "Unknown"
background = NormalizeHex(row.background)
text = NormalizeHex(row.text_color)

IF color is not null:
    frequency[color] += 1
    yearWise[year][color] += 1

Ensure platformAnalysis[platform] exists with counters
platformAnalysis[platform].total += 1
IF color is not null:
    platformAnalysis[platform].colors[color] += 1

result = WCAGResult(background, text, level, textSize)

IF result.status == "pass":
    accessibility.passAA += 1
    platformAnalysis[platform].passAA += 1
ELSE IF result.status == "fail":
    accessibility.failAA += 1
    platformAnalysis[platform].failAA += 1
ELSE:
    accessibility.invalid += 1
```

```
        platformAnalysis[platform].invalid += 1


    accessibility.total += 1


    Ensure accessibilityByYear[year] exists
    Increment year-level pass/fail/invalid counters


RETURN {
    frequency,
    yearWise,
    accessibility,
    accessibilityByYear,
    platformAnalysis,
    wcagCriteria: {
        level: normalized level,
        textSize: normalized textSize,
        threshold: GetWCAGThreshold(level, textSize)  }}
```

4.11.1.6 Flowchart: Accessibility decision

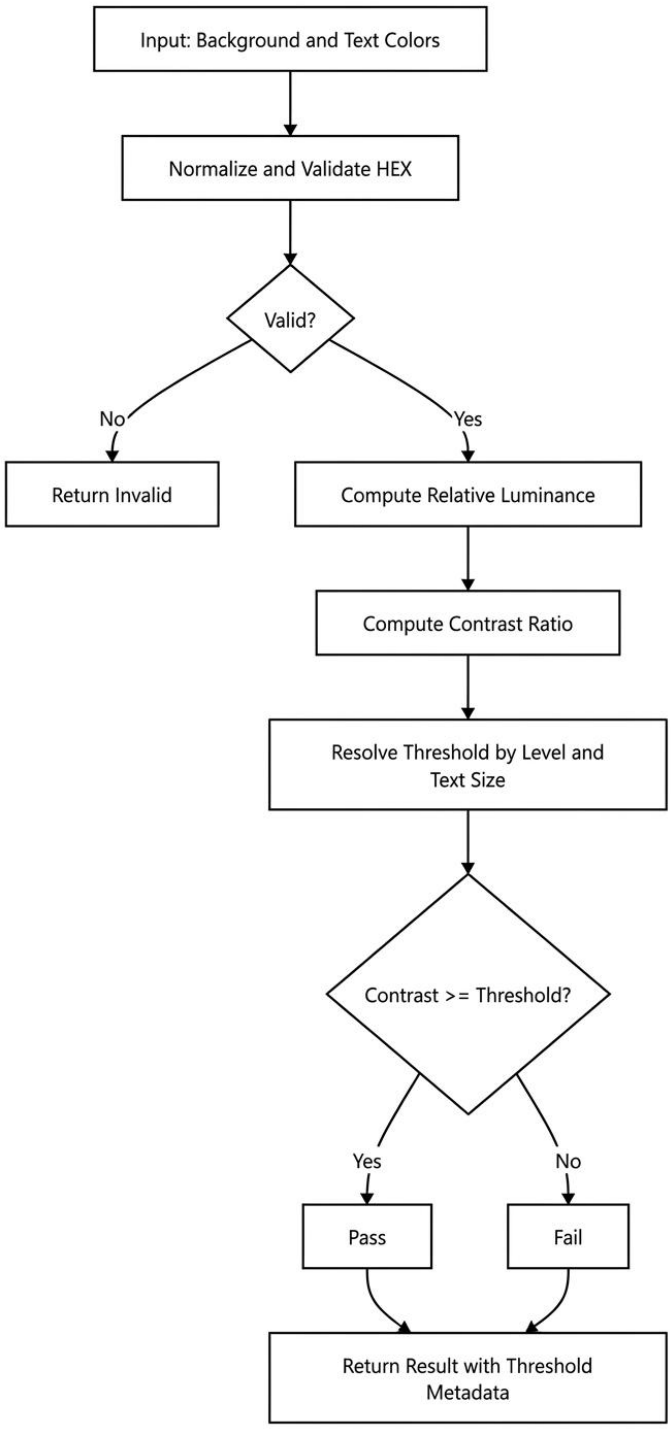


Figure 9 - Flow chart

4.11.2 Framework workflow series

4.11.2.1 Block Diagram A: Data-to-insight pipeline

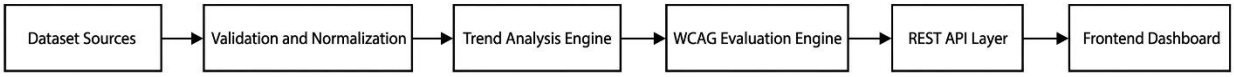


Figure 10 - Block diagram A

4.11.2.2 Block Diagram B: Recommendation interaction loop

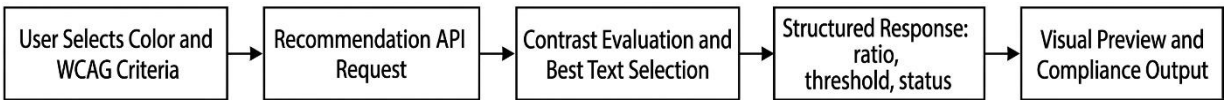


Figure 11 - Block diagram B

4.11.2.3 Block Diagram C: Analysis criteria synchronization

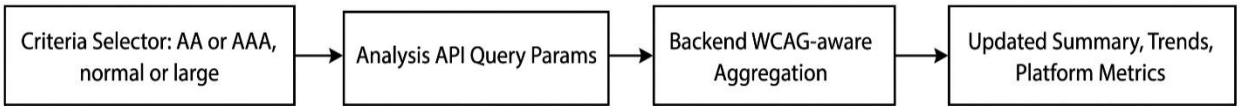


Figure 12 - Block diagram C

4.11.3 Technology selection justification

Table 15 : Technology selection justification

Layer	**Selected Technology**	**Why It Was Selected**	**Contribution to Research Goals**
Runtime	Node.js	Efficient asynchronous I/O and lightweight API hosting	Supports responsive analytics and recommendation services
API Framework	Express.js	Minimal overhead, rapid route composition, middleware ecosystem	Fast implementation of modular REST endpoints
Frontend Library	React	Component-based architecture and state-driven UI rendering	Enables dynamic criterion-driven accessibility visualization
Data Exchange	JSON	Human-readable and portable format	Simplifies dataset persistence and reproducible experiments
HTTP Client	Axios	Stable promise-based API communication	Reliable frontend-backend integration for real-time analysis
Visualization	Recharts	Declarative chart rendering	Clear communication of color trends and accessibility metrics
Testing/Validation	Node test scripts and build validation	Direct verification of route behavior and deployment readiness	Provides execution evidence for correctness claims

4.11.4 Programming language

Table 16 : Programming languages

Language	Role in System	Selection Rationale
JavaScript	Backend logic and frontend application	Single-language full-stack development reduced integration complexity and accelerated iteration

4.11.5 Libraries used

Table 17: Libraries used

Library	Layer	Purpose
Express	Backend	REST endpoint development
Axios	Frontend	API communication
Recharts	Frontend	Pie and trend visualizations
React Router	Frontend	Navigation and page-level modularity

4.11.6 Backend and front-end frameworks used

Table 18 : Backend and front-end frameworks used

Tier	Framework	Key Benefit	Example in Project
Backend	Express.js	Flexible route and middleware handling	Analysis, recommendation, data operations.
Frontend	React	Reusable component architecture	Recommendation and analysis interactive interfaces.

4.12 Testing and evaluation methodology

This part explains how the system was tested and why each testing strategy was selected.

The goal was not to run random tests, but to verify the core research contribution:

- Color trend analysis accuracy.
- WCAG-based accessibility evaluation (AA/AAA, normal/large text).
- Recommendation correctness and consistency between backend and frontend.

To make evaluation meaningful, results were recorded in numeric form (pass rate, response time, error rate) and presented with tables/graphs.

4.12.1 Test plan and test cases

Table 19 :Test plan and test cases

TCID	Test Title	Test Summary	Test Steps	Test Data	Expected Results	Post Condition	Actual Resullts	Status	Notes
TC-01	HEX Normalization Validation.	Verify that valid 3/6-digit HEX values are normalized and invalid values are rejected.	1. Call recommendation endpoint with multiple HEX formats. 2. Observe normalized input in response or error.	#abc, abc, #A1B2C3, ZZZZZZ	#abc/abc normalize to #AABBCC; valid 6-digit remains uppercase; invalid returns error.	Normalization logic confirmed for valid/invalid input paths.	Valid values normalized correctly; invalid input rejected with message.	☑	Confirms preprocessing reliability before WCAG calculations.
TC-	Recommen	Ensure recommenda	1. Send GET request to	/recommend/%23005BAC? level=AA	Response contains input,	Endpoint	API returned	☑	Baseline

02	dation API Basic Success	tion endpoint returns required fields for valid input under AA normal mode.	recommendation API. 2. Check response structure and values.	& textSize=normal (So the full meaning is: "Recommend the best text color for background #005BAC, and check compliance using WCAG AA for normal text.")	suggested Text, contrast, status, level, textSize, threshold.	contract validated for standard mode.	full expected payload with WCAG metadata.		success test for core feature.
TC-03	AAA Normal Strictness Check	Verify stricter AAA-normal threshold behavior.	1. Request recommendation for known color. 2. Compare ratio against AAA threshold.	/recommend/%23005BAC?level=AAA&textSize=normal	6.79:1 should fail against minimum 7:1.	Strict conformance logic validated.	Output showed Fail, threshold: 7, level: AAA, textSize: normal.	☑	Key evidence for dynamic WCAG criteria support.

TC-04	Invalid HEX Error Handling	Confirm invalid color inputs are handled safely and clearly.	1. Send invalid color to recommendation endpoint. 2. Observe HTTP code and error message.	/recommend/INVALIDHEX	HTTP 400 with user-friendly validation error.	No crash; controlled error response generated.	API returned 400 and validation error text.	☑	Prevents corrupted inputs entering analysis flow.
TC-05	Analysis Criteria Switching	Verify /analysis output changes when WCAG criteria are changed.	1. Call /analysis with AA normal. 2. Call /analysis with AAA normal. 3. Compare accessibility counts.	/analysis?level=AA&textSize=normal and /analysis?level=AAA&textSize=normal	Different pass/fail counts under stricter AAA; wcagCriteria returned in both responses.	Criteria-aware aggregation behavior confirmed.	Counts changed as expected; metadata reflected selected mode.	☑	Demonstrates research contribution beyond fixed thresho

									lding.
TC-06	Trend Aggregation Correctness	Validate frequency/year/platform aggregation integrity.	1. Call analysis endpoint. 2. Inspect frequency, yearWise, platform Analysis. 3. Cross-check totals consistency.	Dataset records with primary, year, platform, background, text_color	Aggregated structures populated correctly; totals align with processed records.	Analytical output available for visualization layer.	Aggregations produced expected structures and consistent totals.	☑	Supports trend analytics claims in evaluation chapter.
TC-07	K-Means Clustering Endpoint	Check clustering output validity and format.	1. Call k-means endpoint with k=3. 2. Verify cluster payload and iteration details.	/kmeans?k=3	Returns clusters, centroids, sizes, and iteration count without error.	Cluster model output ready for interpretation.	Endpoint returned valid cluster object and metadata.	☑	Confirms algorithm service availability.

TC-08	API Response-Time Stability	Evaluate repeated-request responsiveness under normal usage.	1. Send 30 repeated requests per key endpoint. 2. Record average and max time.	Endpoints: /analysis, /recommend/%23005BAC	Stable responses; average latency within acceptable lab target.	Performance baseline established for report.	Responses remained stable with no abnormal spikes in test run.	☑	Include numeric chart in thesis.
TC-09	UI and API Output Consistency	Ensure frontend display matches backend-calculated values exactly.	1. Select criteria in UI. 2. Capture UI values. 3. Compare with direct API response for same request.	UI: Recommendation + Analysis pages; API calls with same params	UI status, threshold, level, textSize, and ratio match backend data.	Frontend-backend synchronization verified.	Displayed values matched API outputs in sampled checks.	☑	Important for credibility of visual evaluation results.

TC-10	Production Build Validation	Confirm project compiles successfully for deployment/testing evidence.	1. Run frontend production build. 2. Review compile output.	npm run build in frontend	Build completes successfully without compile errors.	Deployable artifact generated.	Build completed successfully; optimized files generated.	☑	Record build log snippet as execution evidence in appendix.

4.12.2 Nonfunctional testing

Nonfunctional tests were selected to evaluate quality attributes relevant to this research system:

1. Performance: response time for analysis/recommendation endpoints
2. Reliability: repeated request consistency without random output changes
3. Usability: WCAG selector controls correctly update visible results
4. Maintainability/Deployability: successful production build and clean compile status

Suggested numeric metrics

- Average response time (ms)
- 95th percentile response time (ms)
- Error rate (%)
- Build success rate (%)

4.12.3 Functional testing

Functional tests focused on algorithm correctness and business logic:

HEX validation and normalization.

WCAG threshold mapping for AA/AAA and normal/large.

Contrast pass/fail correctness.

Recommendation correctness (best readable option).

Aggregation correctness in trend and accessibility summaries.

API contract validity (required response fields always present).

4.12.4 Testing / Evaluation workflow

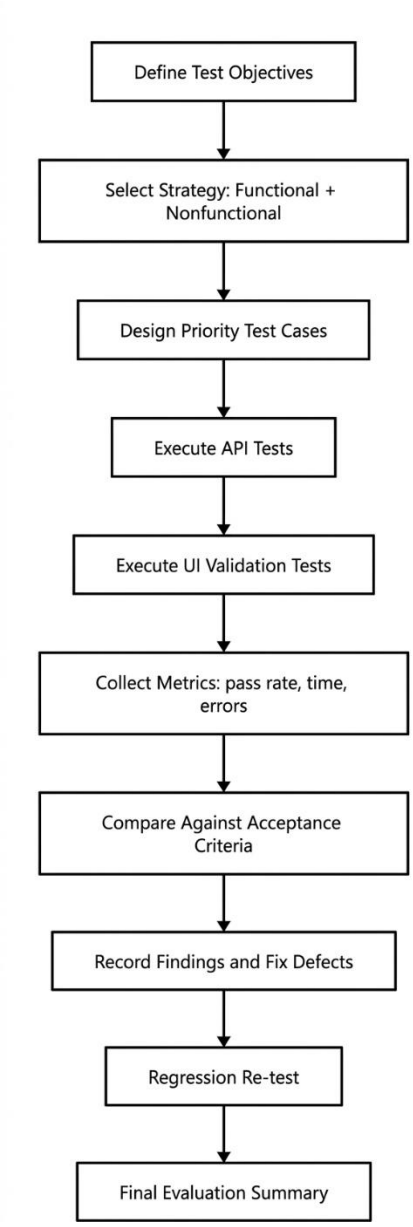


Figure 13 : Testing/evaluation workflow

4.12.5 Review of testing strategies used (research suitability)

The following strategies were chosen because they fit the research objectives:

Black-box testing

Reason: validates API/UI behavior from user perspective without depending on internal code structure.

Boundary and invalid-input testing

Reason: color systems are input-sensitive; malformed HEX and edge values can break evaluation quality.

Criteria-based comparative testing (AA vs AAA, normal vs large)

Reason: directly validates the research contribution of dynamic WCAG evaluation.

Integration testing

Reason: confirms frontend output is synchronized with backend computation and metadata.

Regression testing

Reason: needed after adding new WCAG options to ensure existing trend/recommendation logic still works.

Performance sampling

Reason: confirms the system remains practically usable, not only theoretically correct.

Not all advanced methods are necessary for this project. For example, penetration testing or large-scale load testing is optional unless the research scope includes security hardening or high-concurrency production claims.

4.12.6 Testing methodology summary

The selected testing approach was aligned with the research goals and validated both algorithmic correctness and system behavior. Functional tests confirmed WCAG-aware logic, recommendation validity, and analysis integrity, while nonfunctional tests verified consistency, response quality, and deployment readiness. Numeric outcomes and comparative results provide clear evidence that the implemented system is reliable for research demonstration and future extension.

4.13 Ethical considerations

Ethical considerations are important for maintaining research quality and integrity throughout the study.

The following ethical practices will be followed:

- Using publicly available datasets and resources.
- Providing proper citation for all research materials.
- Avoiding copyright violations.
- Using collected data only for research purposes.

The research will not involve collection of any sensitive personal information from the users.

4.14 Chapter summary

This chapter discussed the methodology employed in the study. This chapter discussed the research design, methods of collecting data, data preprocessing steps, technologies employed, system development methodology, and ethical issues. The methodology gives the framework for the development of the proposed Color Trend Analysis and WCAG-Compliant Color Recommendation System.

5. RESULTS

5.1 Chapter overview

This chapter presents the observable findings and executional evidence produced by the implemented system and its evaluation. The results are presented using tables, figures, algorithm outputs, and test outcomes. Interpretation of what the findings mean is reserved for Chapter 6.

5.2 Dynamic WCAG criteria evaluation results

The implemented WCAG evaluation supports four criteria combinations: AA normal text, AA large text, AAA normal text, and AAA large text.

Executional evidence: the same color pair passed under AA normal with a threshold of 4.5 and failed under AAA normal with a threshold of 7.0. This demonstrates that the output changes according to the selected WCAG criteria.

The implementation evidence includes the WCAG selector controls, active criteria indicator, threshold calculation logic, and pass/fail output.

5.3 Criteria-aware text color recommendation results

For a selected background color, the system calculates contrast against white and black text, selects the higher-contrast candidate, and returns the contrast ratio, selected WCAG level, text size, threshold, and pass/fail status.

Executional evidence: the API returned the expected recommendation fields, and the user-interface output matched the backend response for the same request.

5.4 WCAG-synchronized analysis dashboard results

The analysis dashboard updates accessibility summaries when the WCAG level or text size is changed.

Executional evidence: the pass count decreases when switching from AA normal to the stricter AAA normal mode, and the trend and platform panels reflect the selected criteria consistently.

5.5 Trend and accessibility aggregation results

The integrated analysis process combines color frequency, year-wise trends, platform-wise grouping, and WCAG status counting.

Executional evidence: year-wise accessibility percentages are generated from real dataset rows, and platform comparisons show differences in accessibility performance patterns.

5.6 K-Means clustering results

The K-Means endpoint produced valid cluster objects containing cluster centroids, cluster sizes, member colors, and iteration information for the tested configuration of k=3. The corresponding detailed algorithm and processing logic are described in Chapter 4.

5.7 Functional testing results

The functional evaluation produced successful outcomes for HEX normalization, recommendation API response structure, AAA strictness checking, invalid HEX handling, criteria switching, trend aggregation, K-Means clustering, and frontend-backend output consistency.

Table 20 :Summary of functional testing outcomes

Test Area	**Observed Result**	**Status**
HEX normalization	Valid values normalized correctly; invalid values rejected with a message.	Pass
Recommendation API	Expected recommendation and WCAG metadata returned.	Pass
AAA normal strictness	6.79:1 failed against the 7:1 AAA-normal threshold.	Pass
Invalid HEX handling	HTTP 400 validation response returned without crash.	Pass

Criteria switching	Pass/fail counts changed and selected criteria metadata was returned.	Pass
Trend aggregation	Frequency, year-wise and platform structures were produced with consistent totals.	Pass
K-Means endpoint	Valid cluster object and metadata returned for k=3.	Pass
UI/API consistency	Displayed values matched backend values in sampled checks.	Pass

5.8 Non-functional testing results

The non-functional evaluation considered performance, reliability, usability of WCAG controls, and deployability. Repeated-request testing showed stable responses without abnormal spikes in the tested run. The production build completed successfully and generated optimized files.

The current evidence does not include a verified numerical average or 95th-percentile response-time value; therefore, unsupported numerical values have not been inserted into this thesis. If measured response-time and error-rate figures are available, they should be added here with a chart.

5.9 Production build and deployment validation results

The frontend production build completed successfully without compile errors, providing a deployable artifact for testing evidence.

5.10 Results summary

The results demonstrate successful execution of the major research functions: color normalization and trend aggregation, K-Means clustering, dynamic WCAG evaluation, criteria-aware text-color recommendation, synchronized dashboard analysis, website-related audit capability, and core functional/non-functional validation. The detailed interpretation of these findings is presented in Chapter 6.

5.11 Key implementation attempts (contribution-focused)

This section highlights the parts of the system that directly support the research contribution. This part presents only the implementation attempts that directly contribute to the core research objective of the system: color trend intelligence and WCAG-based accessibility recommendation.

5.11.1 Attempt 1: Dynamic WCAG criteria engine (AA/AAA, normal/large)

One of the most important improvements was moving from a single fixed contrast rule to a flexible WCAG evaluation model. Instead of checking only one condition, the system now supports four practical cases: AA normal text, AA large text, AAA normal text, and AAA large text.

This matters because accessibility is context-dependent. A color pair that passes one level may fail another. By allowing criteria selection at runtime, the system provides more realistic and academically stronger results.

Why this is a contribution

Most simple tools check only one threshold (commonly AA normal). This system extends compliance logic to all major text scenarios, improving practical and research validity.

1.Recommendation page with WCAG selector controls visible -

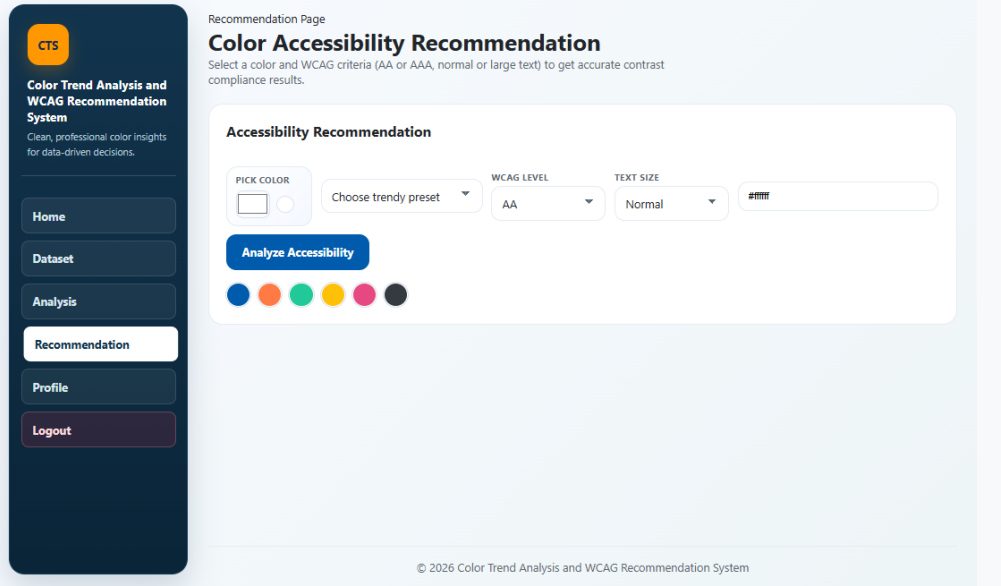


Figure 14 : Recommendation page

2.Criteria badge or label showing active mode

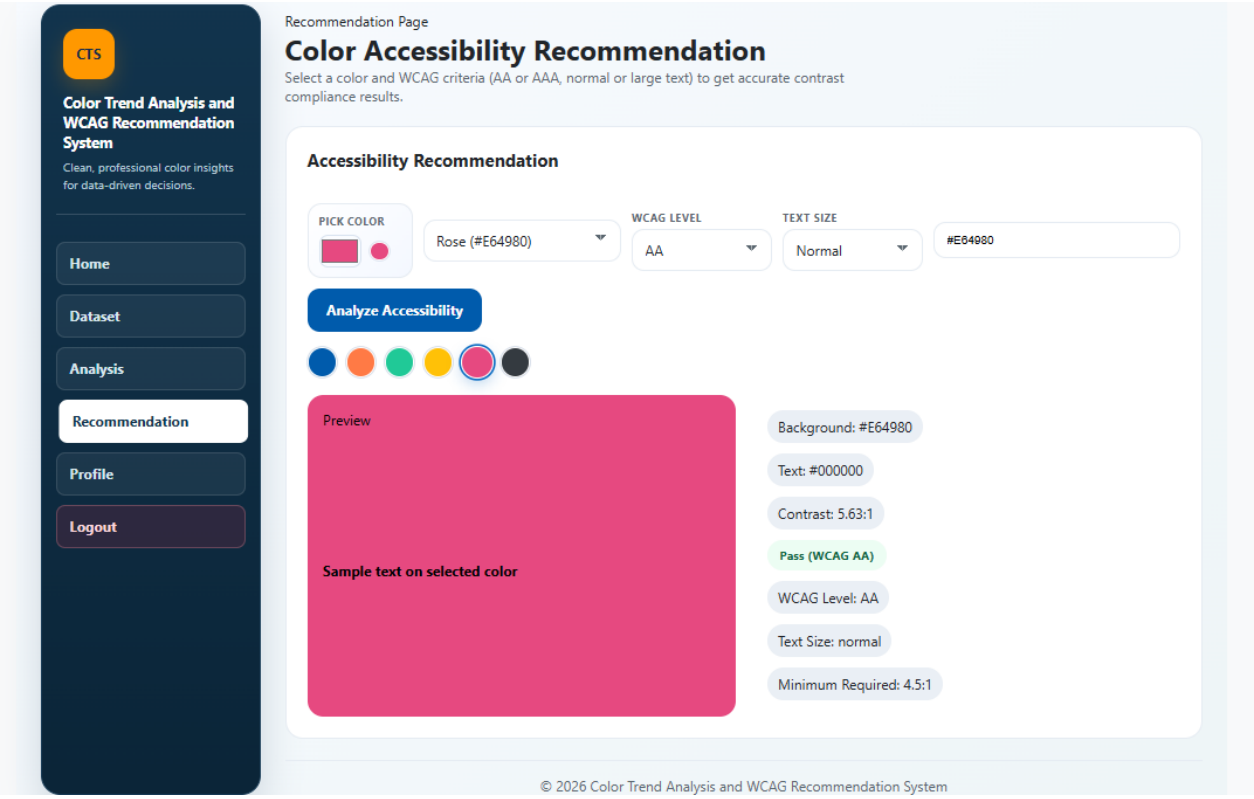


Figure 15 : Recommendation page

Code snippet evidence

threshold = GetWCAGThreshold(level, textSize)

status = (contrast >= threshold) ? "pass" : "fail"

return { contrast, status, threshold, level, textSize }

Executional evidence

1. Same color pair under AA normal: Pass (threshold 4.5).
2. Same color pair under AAA normal: Fail (threshold 7.0).
   This confirms correct criteria-sensitive behavior.

5.11.2 Attempt 2: Criteria-aware text color recommendation

Implementation focus

For a selected background color, the system computes contrast against white and black, then recommends the text color with maximum contrast while also checking pass/fail under the selected WCAG criteria.

Why this is a contribution

The recommendation is not static. It is both optimization-based (best contrast) and standards-aware (contextual compliance), which makes it suitable for real design decisions.

1. **Color picker + preset color selection interface.**

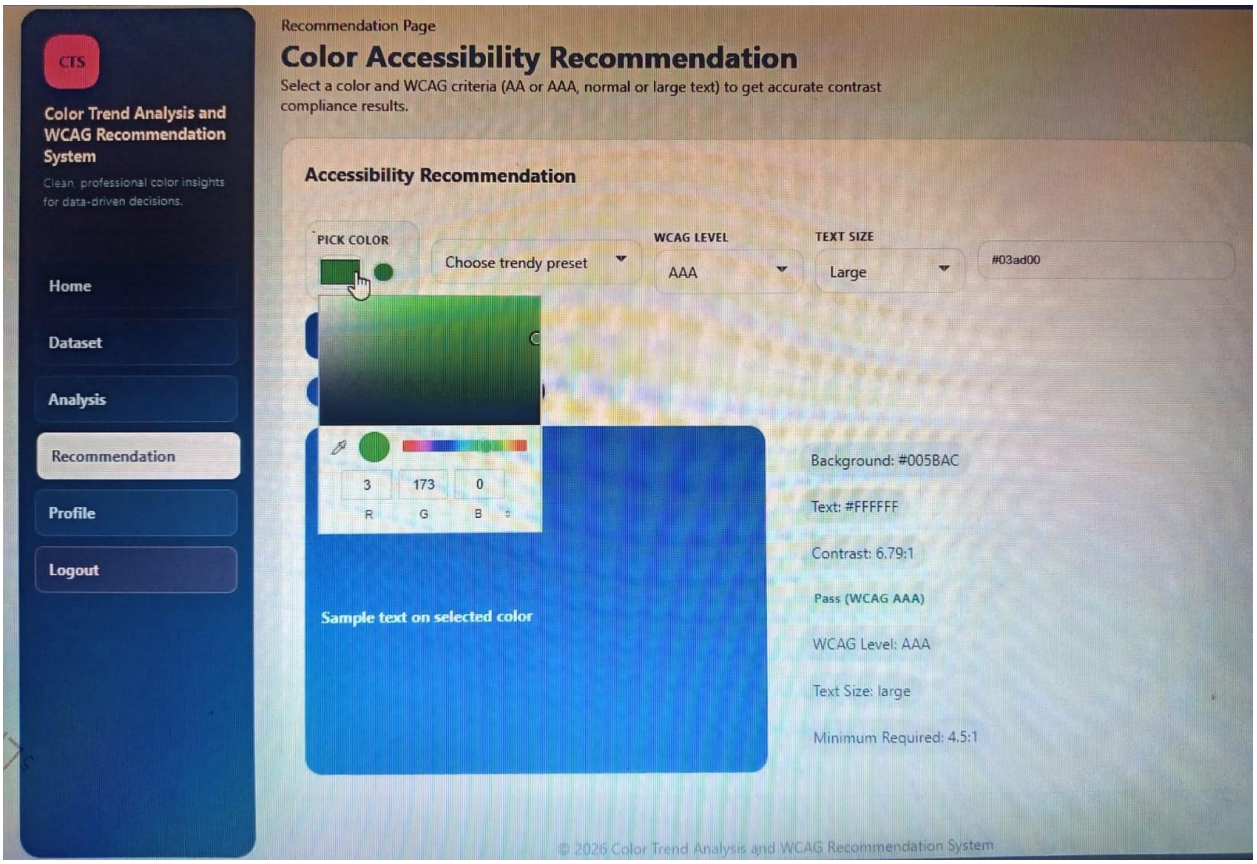


Figure 16 : Color picker

2.Recommendation result card displaying: suggested text, contrast ratio, threshold, pass/fail.

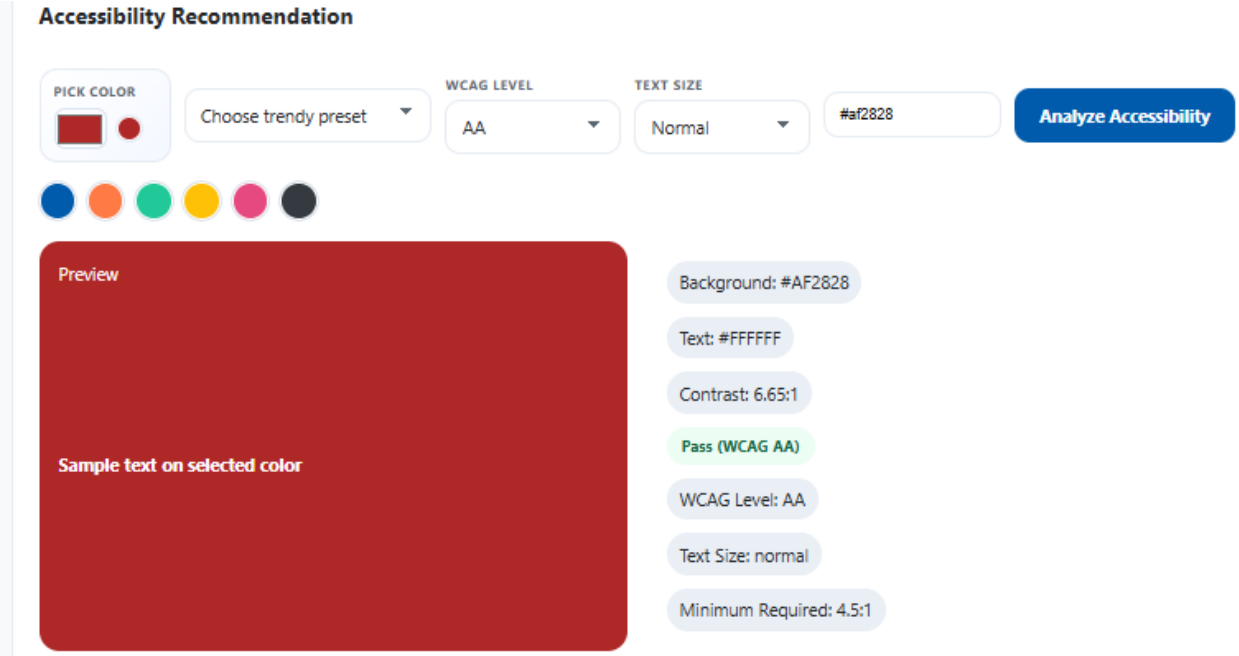


Figure 17 : Recommendation results

Code snippet evidence

cWhite = ContrastRatio(bg, "#FFFFFF")

cBlack = ContrastRatio(bg, "#000000")

suggested = (cWhite > cBlack) ? "#FFFFFF" : "#000000"

best = max(cWhite, cBlack)

status = (best >= threshold) ? "Pass" : "Fail"

Executional evidence

API response includes suggestedText, contrast, level, textSize, threshold, status.

UI output matches backend response for the same request.

5.11.3 Attempt 3: WCAG-synchronized analysis dashboard

Implementation focus

The analysis module was integrated with runtime WCAG controls, enabling aggregated accessibility summaries to update when users switch between AA/AAA and normal/large modes.

Why this is a contribution

This turns the dashboard from static reporting into comparative accessibility analytics. It enables deeper interpretation of how stricter criteria affect pass/fail rates across years and platforms.

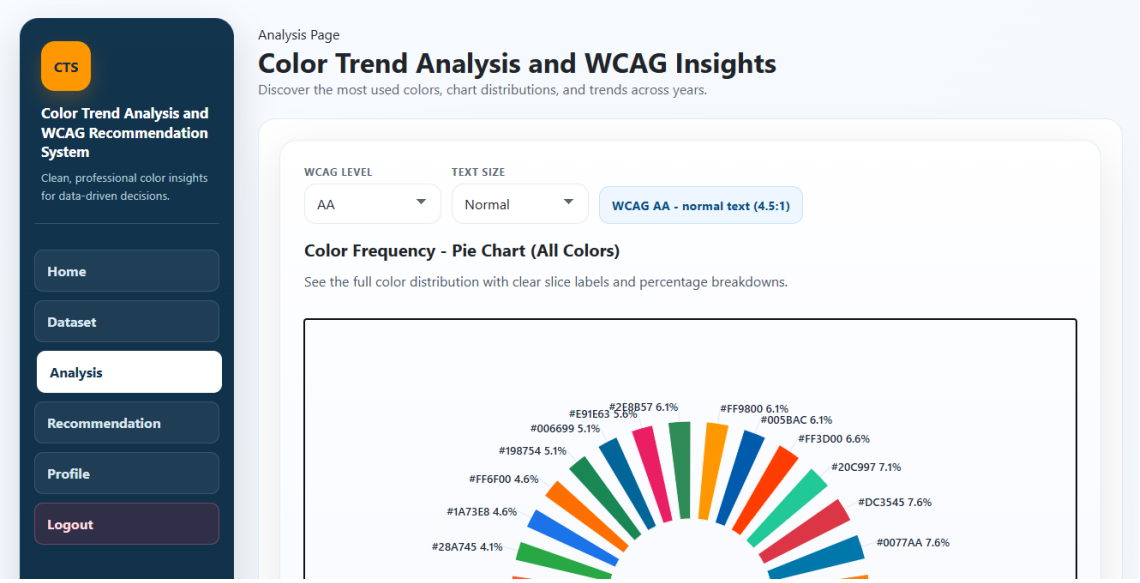


Figure 18 : Analysis page

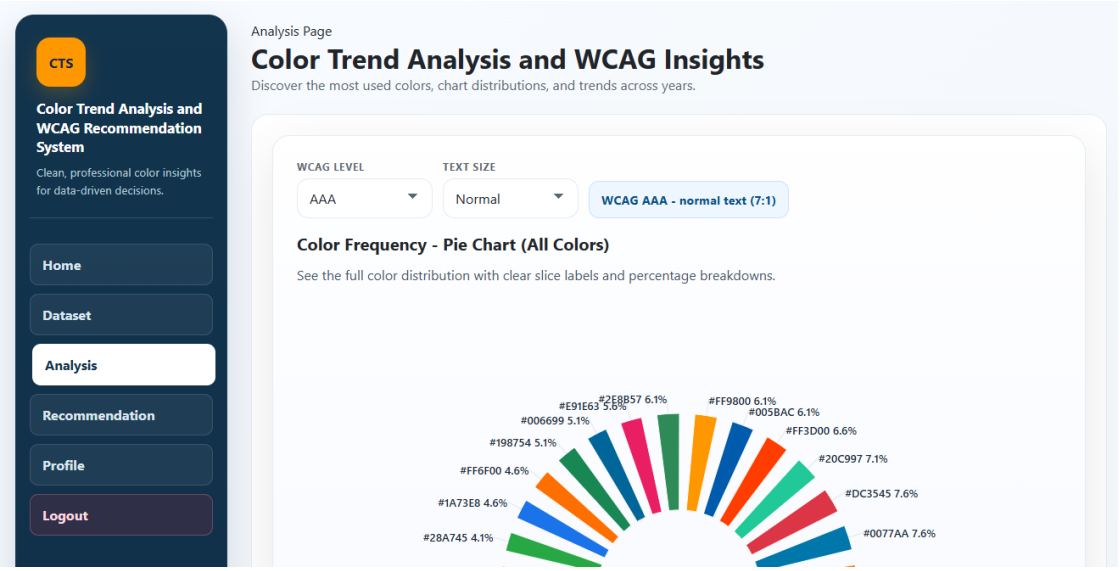


Figure 19 :Analysis page

Accessibility Summary

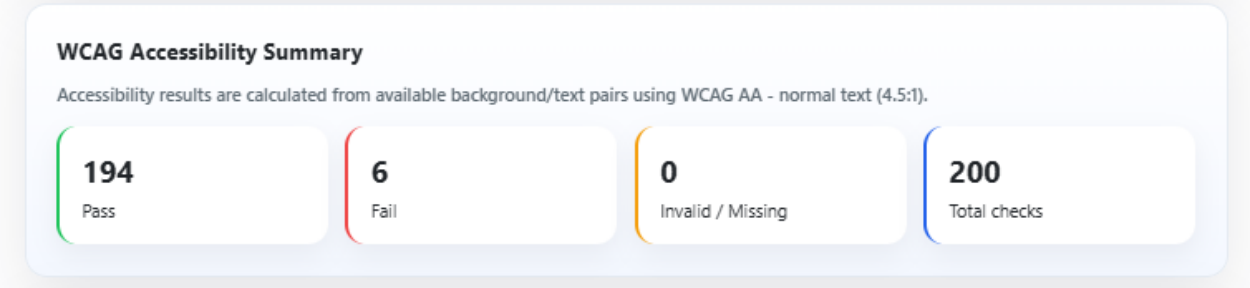


Figure 20:Accessibility page

Code snippet evidence

GET /analysis?level=AAA&textSize=normal

for each row:

result = WCAGResult(background, text, level, textSize)

update pass/fail/invalid counters

return summary + wcagCriteria

Executional evidence

Pass count decreases when switching from AA normal to AAA normal.

Trend and platform panels reflect updated criteria consistently.

5.11.4 Attempt 4: Trend + Accessibility aggregation model

Implementation focus

A unified aggregation process combines frequency analysis, year-wise trend extraction and WCAG status counting in one analytical pipeline.

Why this is a contribution

The model links aesthetic trend data with accessibility performance, which is central to the project's novelty.

1.Color frequency chart (pie chart).



Figure 21 :Color frequency chart

2.Trend vs Accessibility panel.

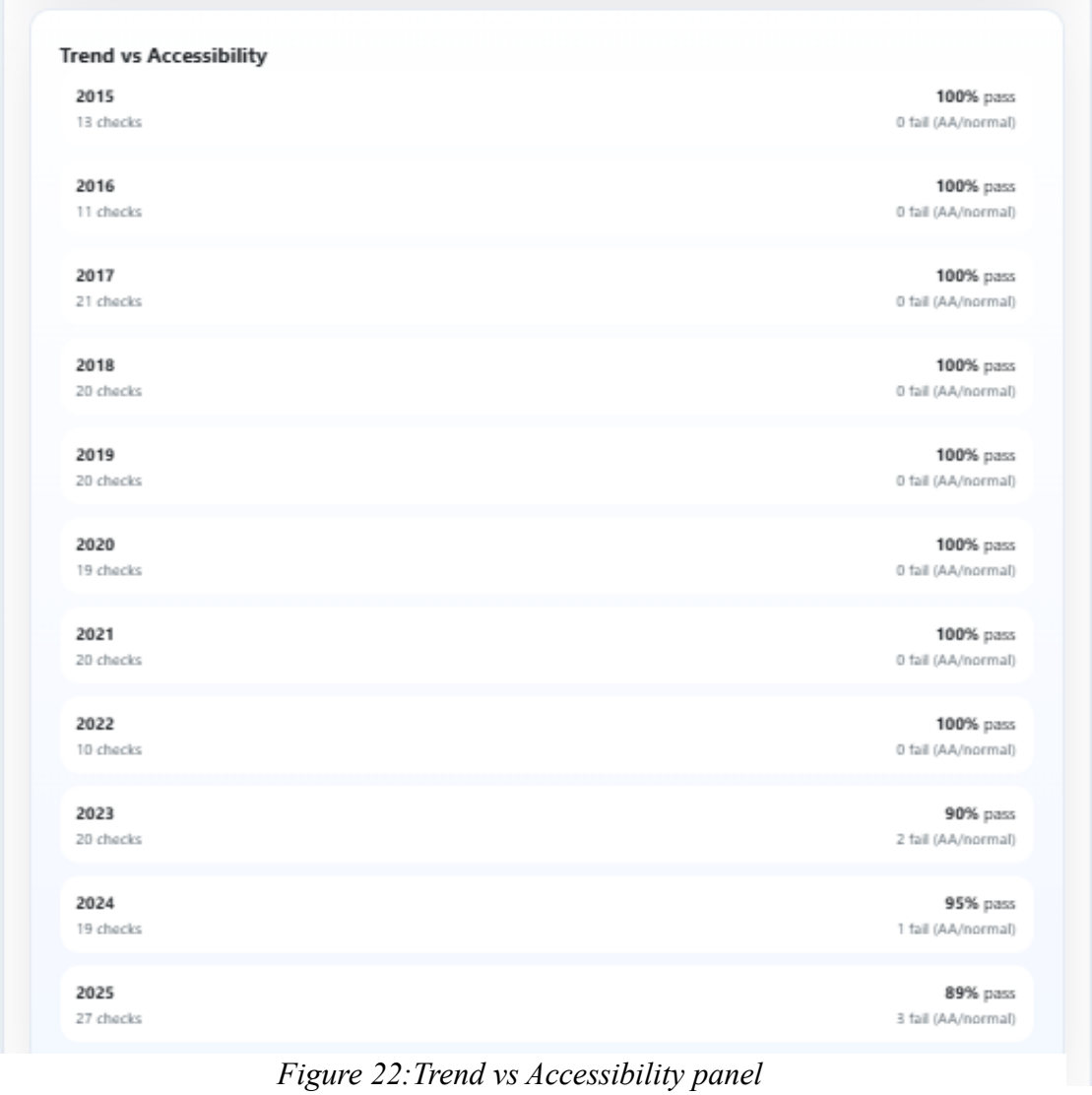


Figure 22:Trend vs Accessibility panel

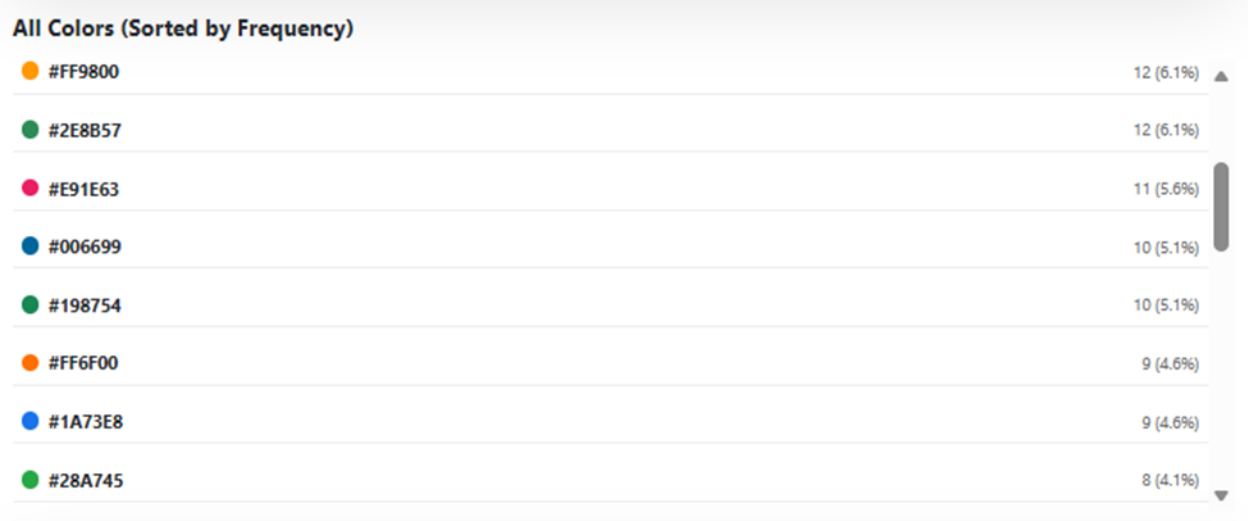


Figure 23:Colors sorted by frequency

Code snippet evidence

for each dataset row:

normalize colors

update frequency and year-wise counts

evaluate WCAG status

update global/year/platform accessibility metrics

return combined analysis object

Executional evidence

Year-wise accessibility percentages are generated from real dataset rows.

Platform comparison reveals different accessibility performance patterns.

5.12 Implementation attempts Summary

This part described how the Color Trend Analysis and WCAG Recommendation System was designed ,implemented from end to end and described the results. The final framework brings together three core parts: color trend analytics, WCAG-based contrast checking, and an interactive recommendation interface built on a React-Express architecture.

The main contribution of this work is the shift from fixed accessibility checking to a dynamic WCAG model. By supporting AA/AAA levels and normal/large text contexts, the system can evaluate accessibility in a more realistic and context-aware way. This makes the results more meaningful than a single-threshold approach.

AA/AAA Decision Chart :

Table 21 : AA/AAA Decision chart

Content Type	**WCAG Level**	**Minimum Contrast Ratio**	**Interpretation**
Normal text	AA	4.5:1	Baseline compliance for readable body text
Normal text	AAA	7.0:1	Enhanced readability standard (stricter)
Large text	AA	3.0:1	Acceptable for headings/large labels
Large text	AAA	4.5:1	Higher readability requirement for large text
UI components and graphics	AA	3.0:1	Required for icons, form controls, and meaningful graphics
UI components and graphics	AAA	3.0:1	Same numeric threshold as AA in WCAG 2.1

The implementation was also supported with execution evidence through API responses, UI behavior, and validation outputs. The results confirm that the system works properly in real use and follows recognized accessibility guidelines. In short, the solution is practical enough to apply and strong enough to support academic discussion, with clear room for future improvement.

6.DISCUSSION AND CONCLUSIONS

6.1 Chapter overview

This chapter interprets the findings presented in Chapter 5 in relation to the research objectives, research questions, literature, practical requirements, and limitations. It then presents the contribution, applications, future recommendations, and conclusions of the study.

6.2 Discussion of color trend and data analysis findings

The integrated trend analysis provides a structured way to examine color frequency, year-wise changes, and platform-level patterns from historical UI/UX data. This supports the research motivation of reducing reliance on isolated subjective selection by providing evidence-based information for design decisions. The findings should be compared with the trends and observations identified in the literature review.

6.3 Discussion of K-means clustering findings

The K-Means component groups normalized RGB color values into clusters and provides representative centroids. This creates an additional analytical view of the color corpus beyond simple frequency counts. The current implementation uses a configurable number of clusters and bounded iterations, making the process deterministic and suitable for repeatable analysis. Future evaluation can compare different values of k and perceptual color spaces.

6.4 Discussion of WCAG accessibility findings

The results show that accessibility evaluation is sensitive to the selected WCAG level and text size. The observed 6.79:1 example passes the AA normal threshold of 4.5:1 but fails the AAA normal threshold of 7:1. This demonstrates why a single fixed threshold does not fully

represent the range of WCAG text-contrast criteria. The system therefore provides a more explicit criteria-aware assessment for UI/UX color decisions.

6.5 Discussion of color recommendation findings

The recommendation mechanism compares white and black text against the selected background and reports the stronger contrast together with the applicable WCAG criteria. Because the rule is explicit and explainable, designers can see why a recommendation was produced rather than receiving an unexplained color suggestion. The approach is intentionally limited and can be extended to brand-aware and palette-based recommendations.

6.6 Discussion of dashboard and website audit findings

The synchronized dashboard links accessibility criteria with aggregated trend and platform information, allowing the same dataset to be examined under different WCAG contexts. The optional website audit extends the analysis from historical dataset records toward practical UI inspection. These features increase the applicability of the system, while their broader generalization should be validated with larger and more diverse website samples.

6.7 Discussion of testing and evaluation findings

The selected testing strategies were aligned with the research contribution. Black-box testing verified observable API and UI behaviour, boundary and invalid-input testing checked input sensitivity, criteria-based testing directly evaluated the dynamic WCAG feature, integration testing checked frontend-backend synchronization, regression testing supported stability after feature changes, and performance sampling addressed practical responsiveness. The successful functional outcomes and production build validation provide evidence that the implemented prototype operates as intended within the tested scope.

6.8 Achievement of research objectives

Table 22 : Achievement of research objectives

Research Objective	**Evidence**	**Discussion**
Objective 1 – Determine present UI/UX color trends	Frequency, year-wise and platform-wise trend analysis results	The system provides a structured data-driven mechanism for identifying color patterns.
Objective 2 – Study the effect of color choice on usability and accessibility	WCAG contrast evaluation and accessibility aggregation	The results demonstrate that contrast outcomes vary with the selected WCAG criteria and therefore affect accessibility assessment.
Objective 3 – Create a WCAG-based color recommendation system	Dynamic WCAG engine and criteria-aware recommendation results	The system produces explainable text-color recommendations together with contrast and criteria metadata.
Objective 4 – Assess system efficiency	Functional, non-functional and production-build testing evidence	The tested functions operated successfully and the production build was validated; further numerical performance measures can strengthen the evaluation.

6.9 Research contribution

The main contribution is an integrated web-based approach that connects historical color trend analysis with WCAG-based accessibility evaluation and explainable color recommendation. Additional contribution-focused features include dynamic AA/AAA and normal/large-text criteria selection, synchronized accessibility analytics, K-Means color grouping, and an optional website color audit.

6.10 Concluding remarks

6.10.1 Accomplishment of the research objectives

This research set out to build a system that can do three things in one workflow: analyze color trends, evaluate accessibility using WCAG rules, and provide practical text-color recommendations. These objectives were achieved through an integrated implementation and evaluation process.

To justify this properly, a triangulation-style view can be used from testing outcomes:

Technical triangulation: backend API outputs, frontend UI outputs, and algorithm-level logic gave matching results.

Criteria triangulation: the same inputs were tested under AA/AAA and normal/large text contexts to confirm dynamic WCAG behavior.

Evidence triangulation: functional tests, nonfunctional checks and build/execution logs collectively confirmed reliability.

So, the final outcome is not only a working prototype, but a validated system aligned with accessibility standards and research goals.

6.10.2 Problems encountered

Several practical issues appeared during development and testing:

Data consistency issues: color values came in different formats, requiring strict HEX normalization.

Fixed-threshold limitation at early stage: initial WCAG checking was too static and had to be redesigned for dynamic AA/AAA modes.

Frontend-backend synchronization: criteria shown in UI had to exactly match server-side calculations.

Environment and startup instability at times: build and run behavior needed repeated verification to ensure stable execution evidence.

Scope management challenge: balancing research contribution features while avoiding over-expansion into non-core modules.

These problems were useful because each one pushed the system toward a clearer and more rigorous final design.

6.10.3 Limitations

The study is limited by the scope and representativeness of the available historical secondary dataset. The current recommendation strategy mainly compares black and white text rather than generating complete brand-aware palettes. K-Means clustering uses RGB distance, which does not fully represent perceptual color similarity. The current evaluation is primarily system-level and does not yet provide a large-scale controlled usability study with professional UI/UX designers. Performance evidence should also be expanded with larger benchmark workloads before making high-concurrency production claims.

6.10.4 Self-reflection

6.10.4.1 Ideology about the research carried out

This work represents the idea that design quality should not be separated from accessibility quality. Color can be trendy ,but if it fails readability standards and its real-world usability is limited. The research therefore treated aesthetics and accessibility as co-equal decision factors, not competing ones.

6.10.4.2 Benefits gained

The research delivered multiple practical and academic benefits:

1. A reusable WCAG-aware recommendation workflow.
2. A clearer method to evaluate color choices under different accessibility strictness levels.
3. A full-stack implementation experience from algorithm design to UI evidence presentation.
4. Stronger confidence in presenting results through measurable testing rather than assumptions.

6.10.4.3 Learning curves

Key learning curves during the project included:

1. Translating standards (WCAG text) into correct computational logic.
2. Designing test cases that prove contribution, not just basic functionality.
3. Maintaining consistency between what the API computes and what the interface displays.
4. Structuring research writing so implementation details and evaluation evidence remain logically connected.

6.10.5 Business insight of the idea

The concept has business relevance because accessibility compliance is increasingly expected in digital products. Organizations often struggle to balance brand identity with readability standards. This system provides quick, explainable support for that decision.

Potential value points:

- Faster design review cycles.
- Early detection of non-compliant color choices.
- Better inclusivity and user trust.
- Reduced rework cost during UI refinement.

6.10.5.1 Real-world application possibilities

The idea can be applied in varity of practical contexts:

- Design teams checking color palettes before release.
- Product teams validating dashboard and reporting interfaces.
- EdTech and e-government portals where accessibility is critical.
- Brand systems that need WCAG-aware theme selection.
- Internal compliance tools for UI/UX quality gates.

6.10.6 Future recommendations

The current system provides a strong base, but several meaningful extensions are possible:

1. Expand recommendation candidates beyond black/white into brand-aware color palettes.
2. Add automated “closest compliant color” suggestions when selected combinations fail strict modes.
3. Introduce larger benchmark datasets across industries for stronger trend generalization.
4. Add exportable compliance reports for audit and documentation use.

5. Include deeper performance and usability testing with real users.
6. Explore advanced color-space methods for clustering and perceptual similarity.
7. Consider optional AI-assisted palette generation while preserving WCAG explainability.

6.11 Conclusion

This research developed and evaluated a web-based system that combines color trend analysis, WCAG-based accessibility evaluation, clustering, and criteria-aware text-color recommendation for UI/UX design. The implementation and evaluation evidence show that the proposed workflow can transform historical color data and color inputs into structured trend, accessibility, and recommendation outputs. The dynamic WCAG model provides explicit assessment across AA/AAA and normal/large-text scenarios, while the testing results demonstrate successful operation of the main functional components and production build. The study therefore provides a practical foundation for data-driven and accessibility-aware color decision support, while recognizing the need for larger datasets, broader recommendation strategies, perceptual clustering, and real-user evaluation in future work.

6.12 Chapter summary

This chapter interpreted the results, linked them to the research objectives and literature, discussed the contribution and limitations, identified practical applications and future improvements, and presented the overall conclusion of the study.

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Appendix – A : Additional System Diagrams

Deployment diagram

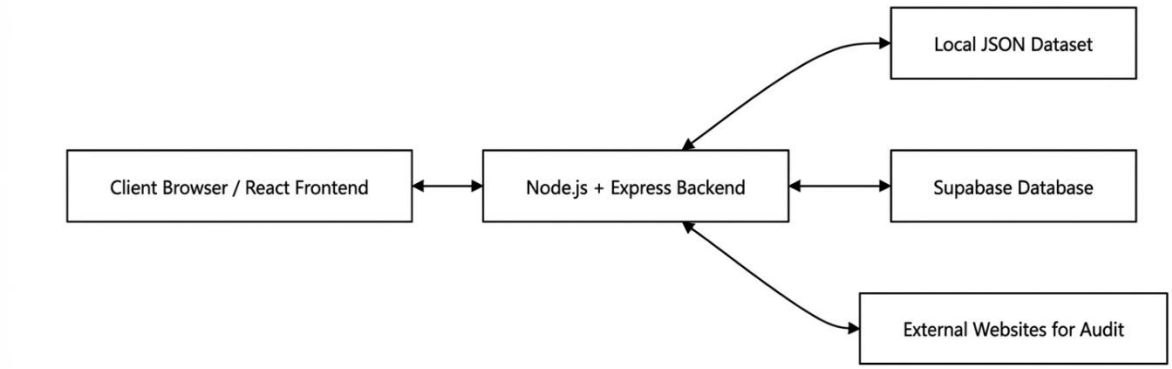


Figure 24 :Deployment Diagram

State chart

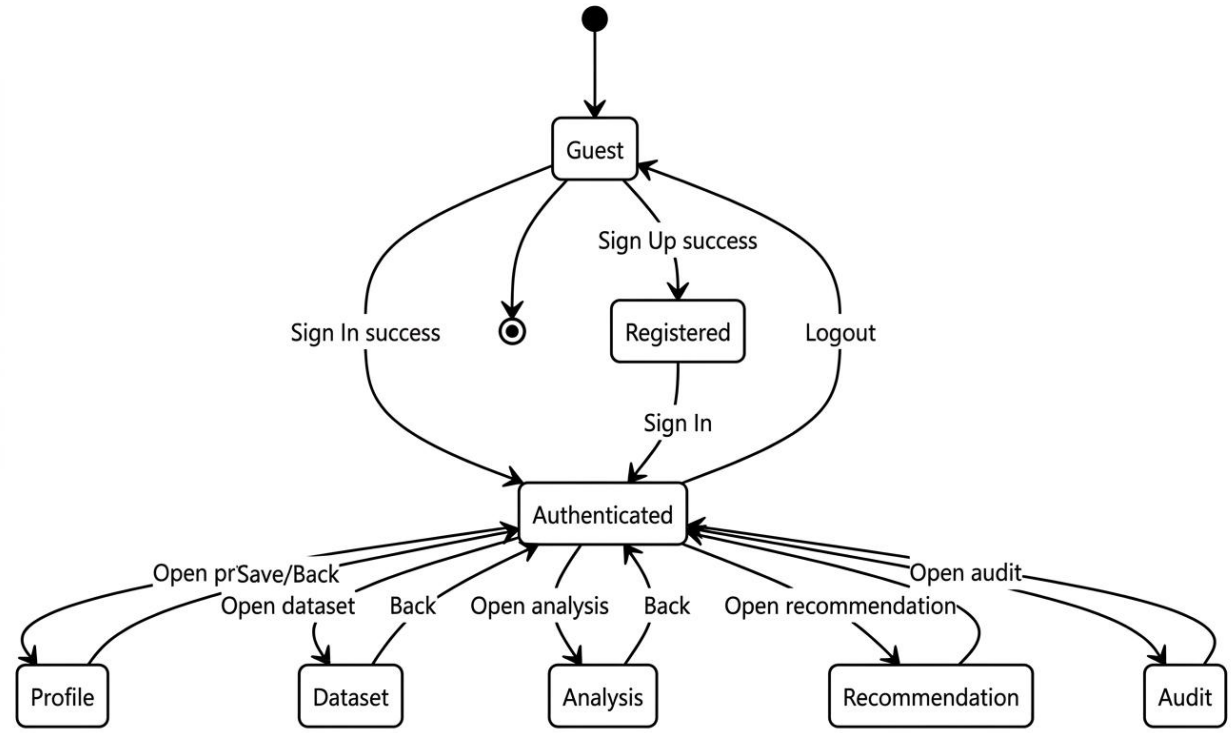


Figure 25 : State Chart

Appendix – B : System Screenshots

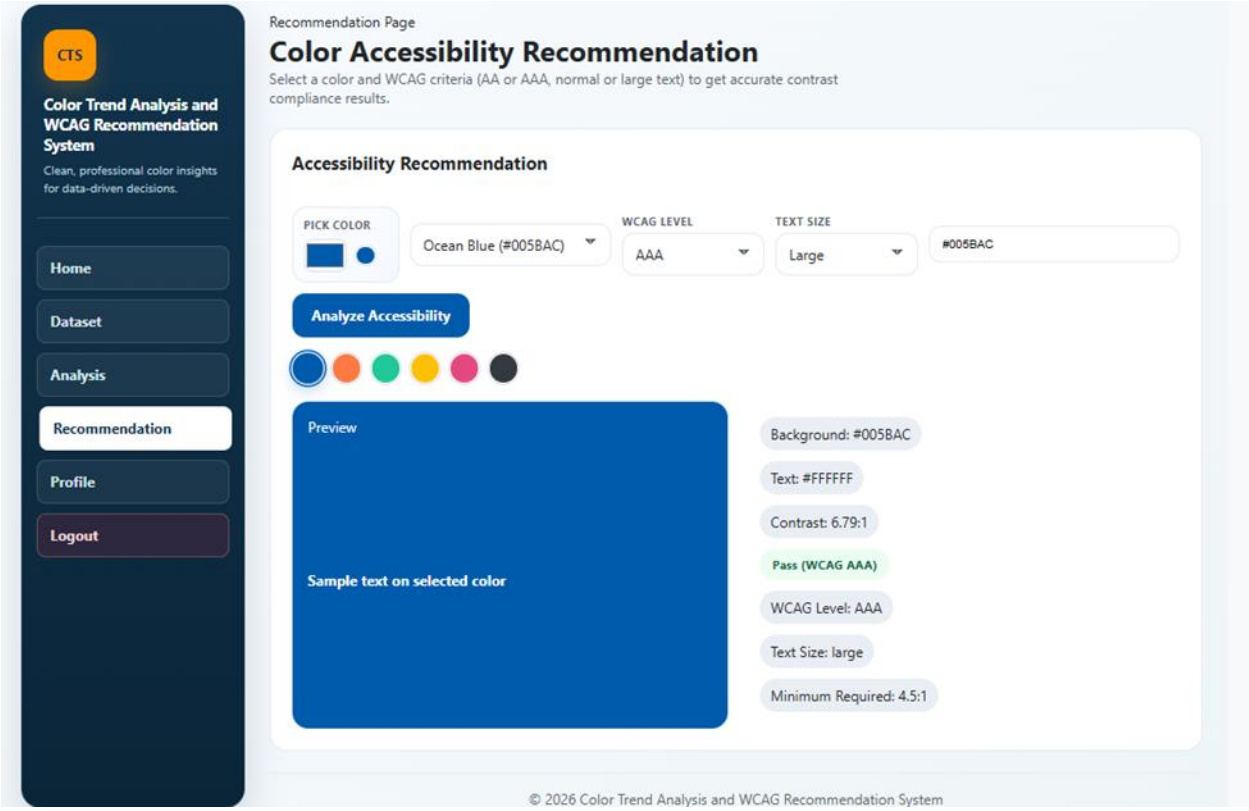


Figure 26 :system screenshot

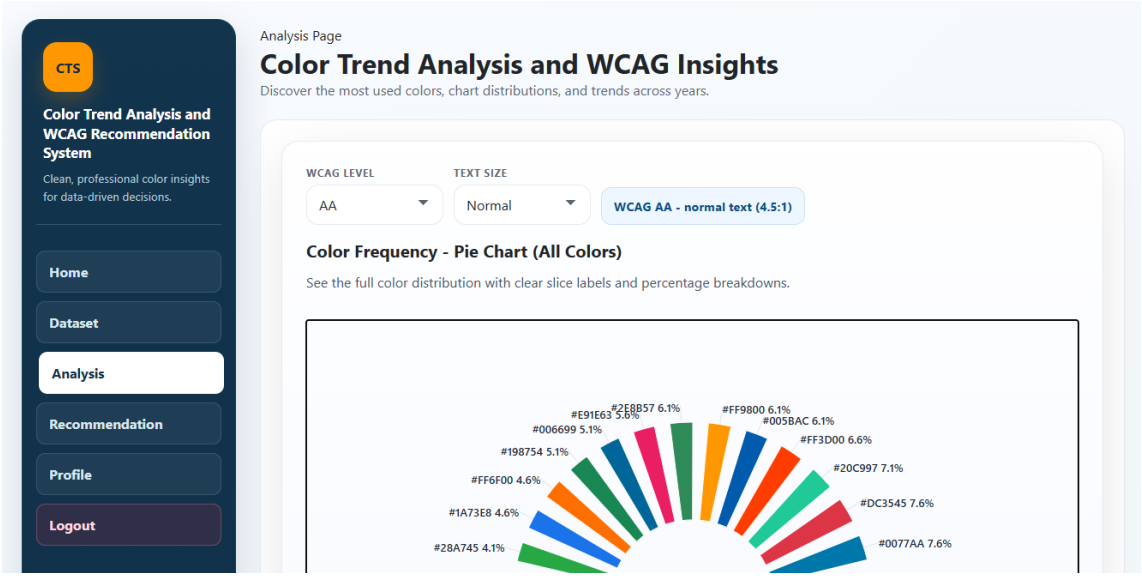


Figure 27 : System screenshot

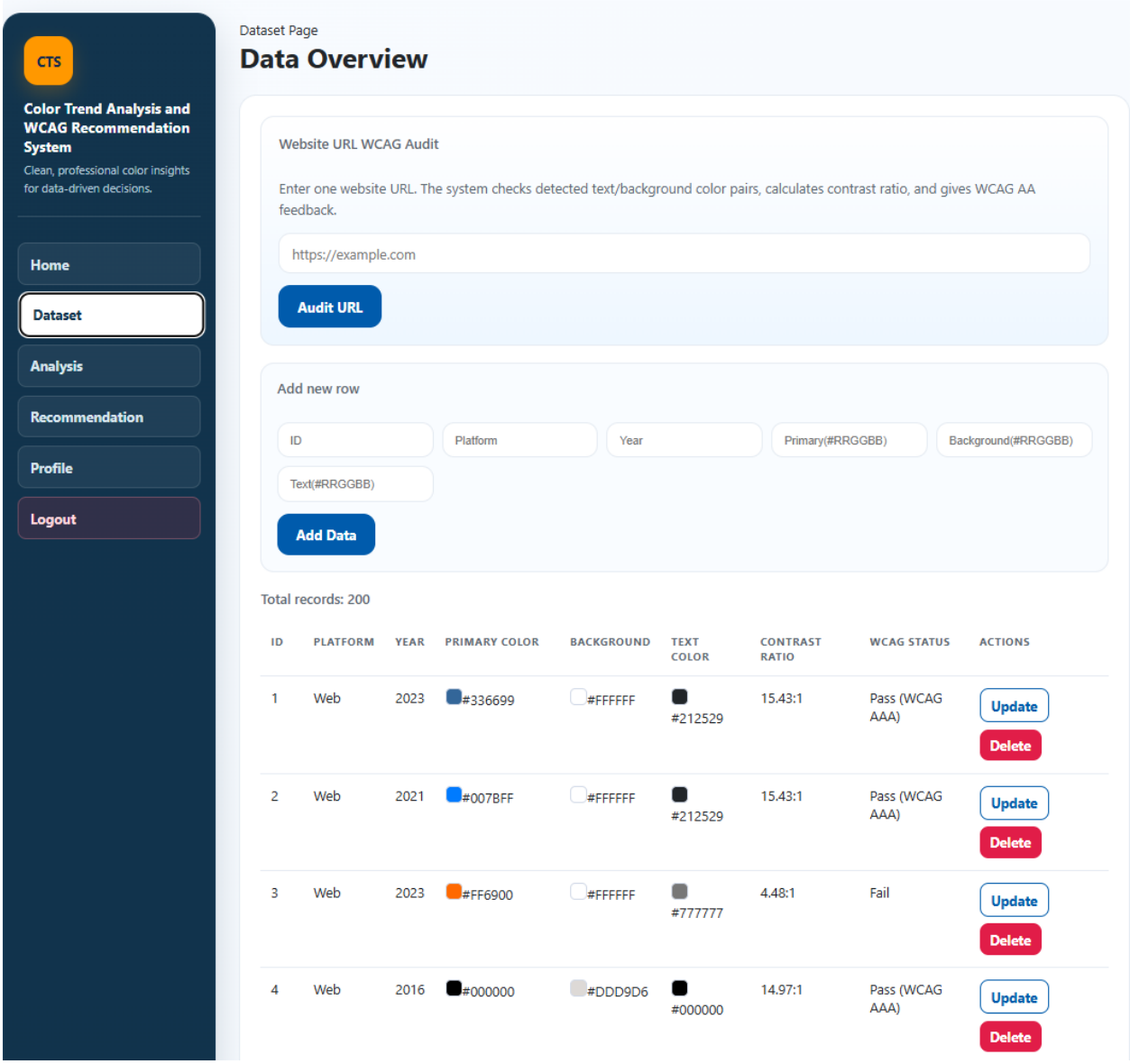
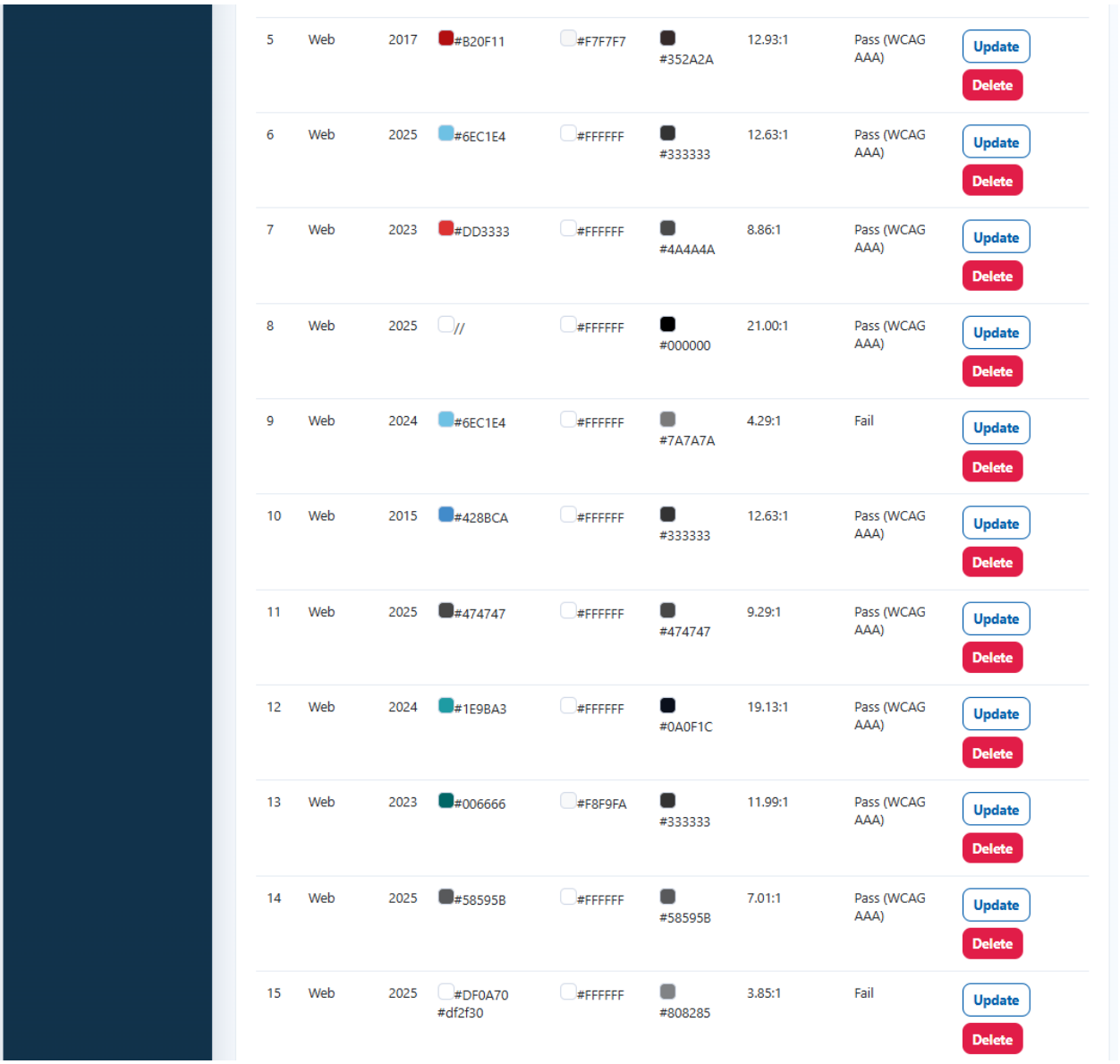


Figure 28 : Dataset page



5	Web	2017	#B20F11	#F7F7F7	#352A2A	12.93:1	Pass (WCAG AAA)	Update Delete
6	Web	2025	#6EC1E4	#FFFFFF	#333333	12.63:1	Pass (WCAG AAA)	Update Delete
7	Web	2023	#DD3333	#FFFFFF	#4A4A4A	8.86:1	Pass (WCAG AAA)	Update Delete
8	Web	2025	//	#FFFFFF	#000000	21.00:1	Pass (WCAG AAA)	Update Delete
9	Web	2024	#6EC1E4	#FFFFFF	#7A7A7A	4.29:1	Fail	Update Delete
10	Web	2015	#428BCA	#FFFFFF	#333333	12.63:1	Pass (WCAG AAA)	Update Delete
11	Web	2025	#474747	#FFFFFF	#474747	9.29:1	Pass (WCAG AAA)	Update Delete
12	Web	2024	#1E9BA3	#FFFFFF	#0A0F1C	19.13:1	Pass (WCAG AAA)	Update Delete
13	Web	2023	#006666	#F8F9FA	#333333	11.99:1	Pass (WCAG AAA)	Update Delete
14	Web	2025	#58595B	#FFFFFF	#58595B	7.01:1	Pass (WCAG AAA)	Update Delete
15	Web	2025	#DF0A70 #df2f30	#FFFFFF	#808285	3.85:1	Fail	Update Delete

Figure 29 : Dataset page

Audit URL

Check the - https://www.atlas.lk/

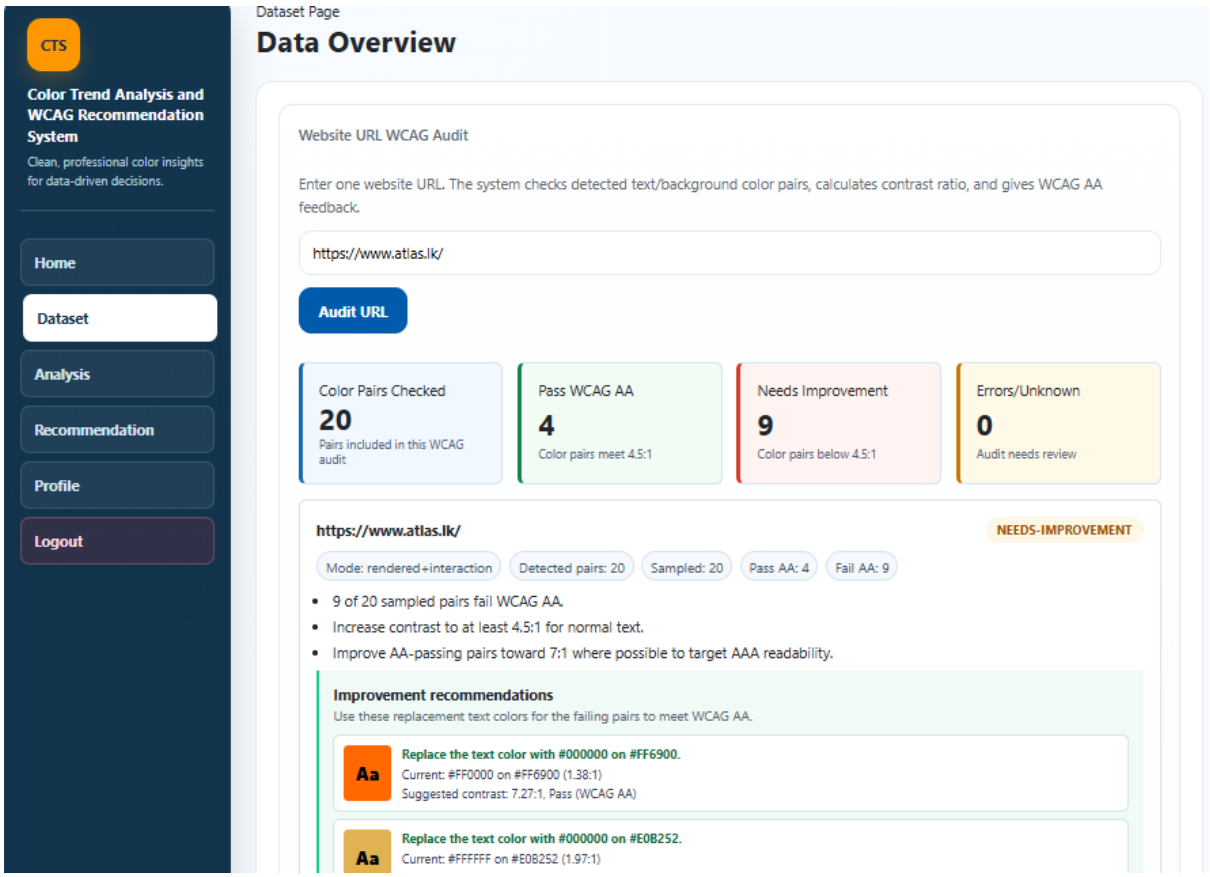


Figure 30 : Audit url

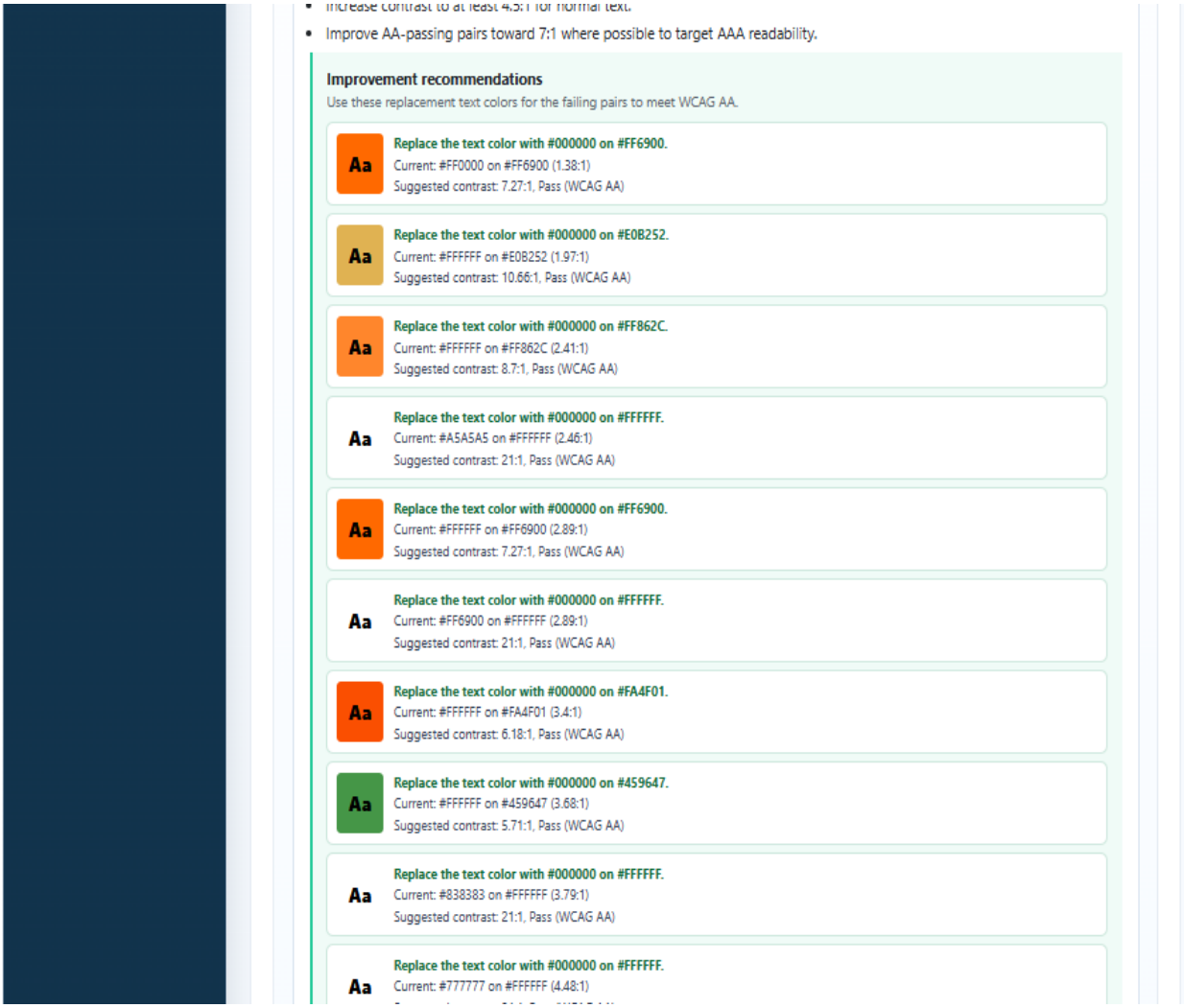


Figure 31 : Improvement and recommendation (audit url)

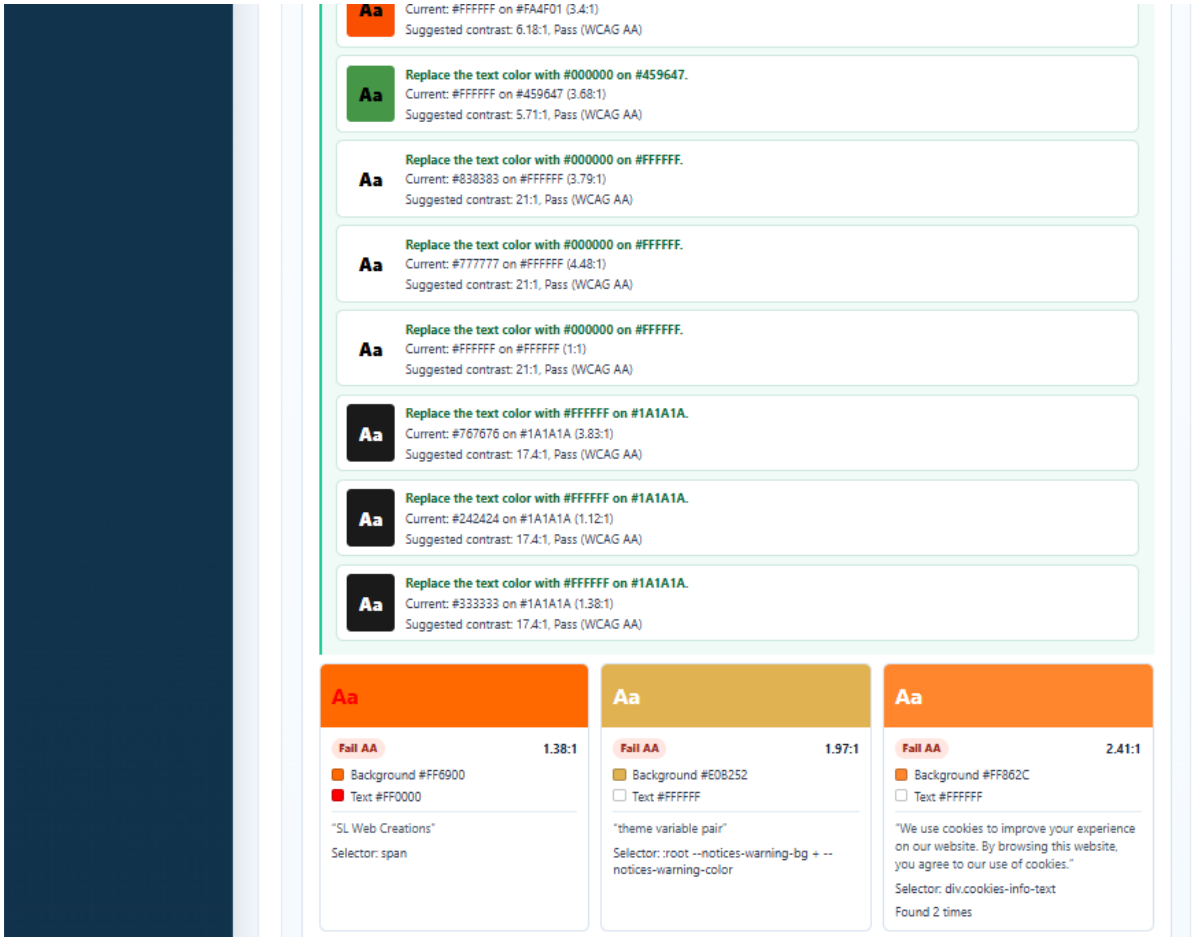


Figure 32 : Output of audit url

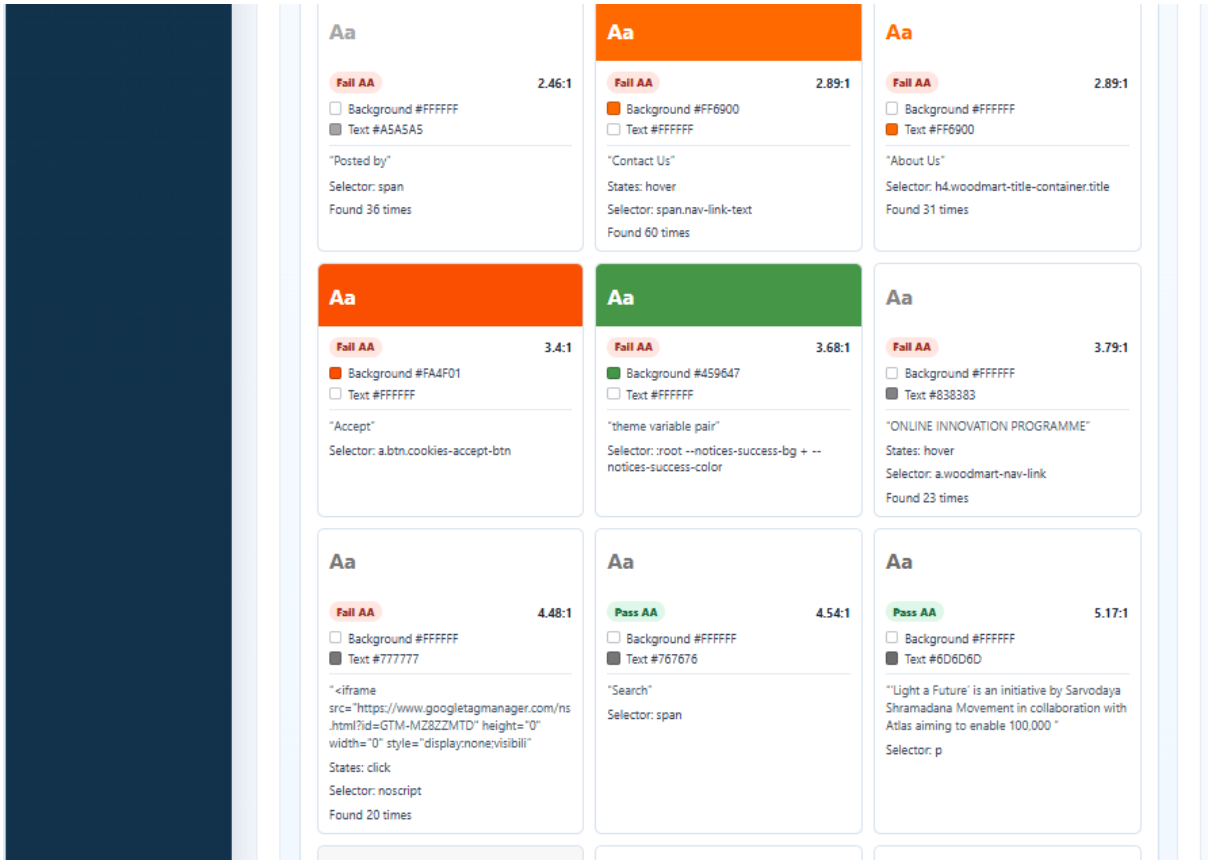


Figure 33 : Output of audit url

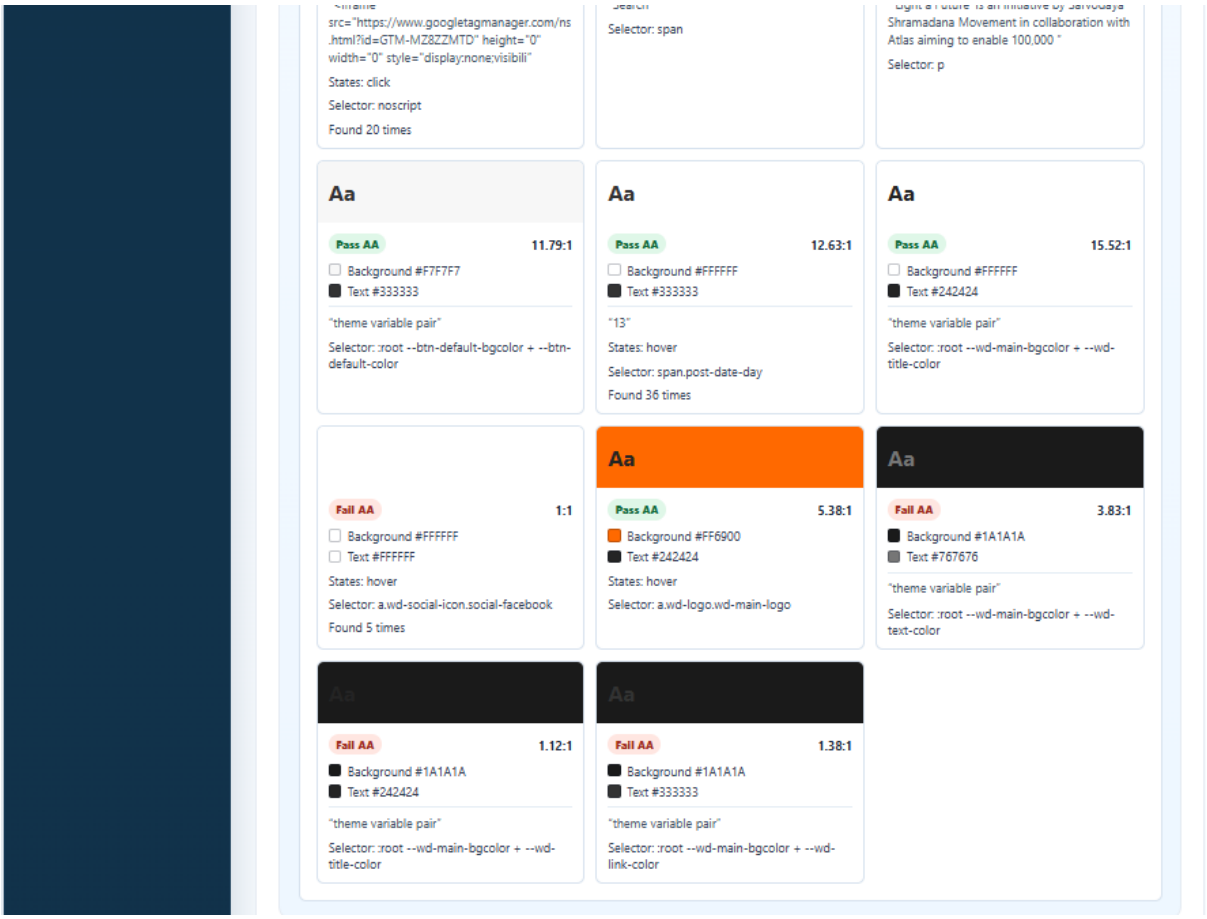


Figure 34 : Output of audit url

Appendix – C : Supervisory meeting minutes

The next pages contain the meeting minutes held with the research supervisor in the process of doing the research. The meeting minutes prove that progress has been made in the course of the research. The meetings assisted in the reviewing of the progress made in the research process, solving the problems connected with the research, assessing the progress in developing the proposed system, and finding out what more needed to be done. The pages below provide supporting evidence for the discussion of the supervisor and progress in the research project.